

SRI LANKAN LEGAL INFORMATION RETRIEVAL SYSTEM

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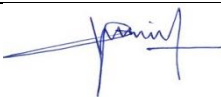
Sri Lanka

September 2023

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ABSTRACT

The Sri Lankan Legal Information Retrieval System is a comprehensive system designed to assist legal professionals in Sri Lanka in accessing legal information quickly and efficiently. The system comprises several stages, including data collection, pre-processing, cleaning, structuring, key information extraction, legal document summarization, and legal decision support development.

The system's primary objective is to provide legal professionals with a user-friendly and efficient system to access legal information and make informed decisions based on legal documents and case law. The system aims to address the challenges faced by legal professionals in Sri Lanka in accessing and analyzing vast amounts of legal data.

The data pre-processing and cleaning stage ensures that the data is in a standardized format. Structured data is then used to develop a legal knowledge graph, which facilitates efficient and effective retrieval of legal documents. The key information extraction and legal document summarization component provides legal professionals with a concise overview of the most critical legal concepts and arguments in a document. The legal decision support system provides legal professionals with assistance in making informed decisions by answering legal queries based on legal documents.

The Sri Lankan Legal Information Retrieval System is a critical tool for legal professionals in Sri Lanka, providing efficient access to legal information and assisting in making informed decisions based on legal documents and case law.

Key Words: Legal Information Retrieval, Legal Decision Support, Natural Language Processing, Legal Knowledge Graphs, Legal Document Summarization

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LIST OF ABBREVIATIONS

Abbreviation	Definition
FOC	Faculty of Computing
NLP	Natural Language Processing
NLTK	Natural Language Tool Kit
CDAP	Comprehensive Design and Analysis Project
OCR	Optical Character Recognition
SLIIT	Sri Lanka Institute of Information Technology
OCR	Optical Character Recognition
KG	Knowledge Graph
API	Application Programming Interface
ML	Machine Learning
GNN	Graph Neural Network
NER	Named Entity Recognition
NLR	New Law Reports
SLR	Sri Lankan Law Reports
IR	Information Retrieval
BERT	Bidirectional Encoder Representations from Transformers
MOJ	Ministry of Justice
NLP	Natural Language Processing
SRS	Software Requirements Specification
RoI	Return on Investment
IDE	Integrated Development Environments
HTTP	Hyper Text Transfer Protocol

1 INTRODUCTION

1.1 Background & Literature Survey

The law could be defined as the backbone of modern society. Humans have created the rule of law to bring order to the chaotic world [1]. The legal framework is the structure that makes human society a civilized one [2]. Legal systems and the legal framework have evolved throughout civilization all around the world. With the modernization of technology, the evolution of the legal system has also happened rapidly and drastically.

The legal domain is vastly dependent on documentation and information. As stated above it is evident that technological modernization has affected the entire world, thus, it has directly affected the legal domain too. Due to the advancements in technology, it has become very easy to create and output legal documents and information. Therefore, the legal systems have managed to produce a plethora of information in various forms of documents in recent decades. The above has created a counterproductive aspect of information utilization. Due to the vast amount of available information, information retrieval and processing have become extremely difficult. This has been discussed in an article in the Harvard Business Review by the author Paul Hemp. It has become evident that, with technological advancements, information availability has become very abundant. This plethora of information has surpassed the human brain's capacity to comprehend and absorb information within a person's lifetime [3].

The above phenomenon is a universal issue. When we look at the Sri Lankan context a study has been conducted by the author Aquinas V. Tambimuttu about the Sri Lankan legal system. In this study, the author highlights six main types of courts in Sri Lanka: the Supreme Court, the Court of Appeal, the High Court, the District courts, the Magistrate's courts, and the Primary courts [4] [5]. These court systems have been established to enforce the laws created by the Parliament of Sri Lanka. The Lankan legal system is a mix of several laws and is unique to the country of Sri Lanka. It is a concoction of English common law, Roman-Dutch Law, Kandyan Law, Thesawalamai Law, and Muslim Law [4]. Thus, it is evident that the mixture of laws and the legal infrastructure of the country

produces a vast volume of legal information constantly, which redirects to the problem highlighted above. Thus, the need for a proper legal information retrieval system could be identified.

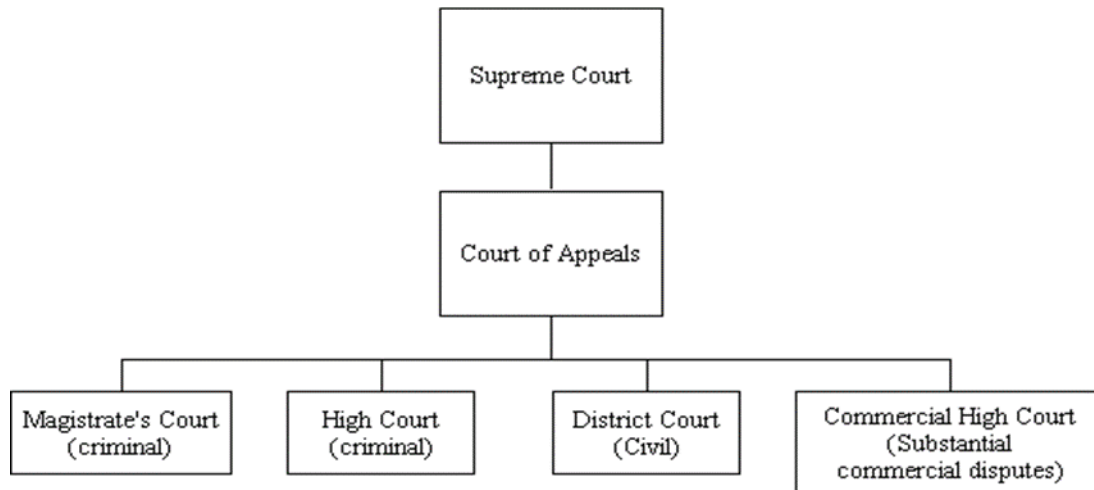


Figure 1 Court structure of Sri Lanka

Source: University of Minnesota, Human Rights Library

The inability, lack of access, and lack of legal knowledge is the main problem for socio-cultural issues. Through a study done by the authors R. O. Moorthage and H. P. K. N. Hewawasam in the Rathnapura Police Division it has been uncovered, the impact of legal literacy on crime control and prevention. Through this study, the authors have discovered that citizens with the least knowledge about the legal framework, rights, and punishments are the ones who are mostly involved in illicit activities. Ignorance of the rule of law is also a major cause of becoming victims of illegal activities [2]. Thus, it is evident that having proper access to legal knowledge could be beneficial for a civilized society.

The legal domain and access to legal information have become tedious tasks for legal professionals and law students too. These parties are highly dependent on the management of the ever-increasing legal knowledge base, and it affects the productivity and efficiency of these personnel. The importance of legal information management and its effects have been studied by the authors Nina Evans and James Price. Through their studies, it is

evident that the proper management of legal information is very important, and this would enable legal professionals to uphold the law to a higher standard [6]. Another study by the Sri Lankan author H.M.P.P Karunarathna has identified that a majority of law students use electronic means of accessing legal information, but out of this, a vast majority suffer due to the unavailability of consistent, accurate, and proper legal information systems [7]. Thus, it is evident that there is an issue in legal information management, information retrieval, and the availability of a proper legal knowledge base.

The disproportionate availability of legal information is a big problem in the country. The Sri Lankan legal system contains legal information such as Acts of Parliament, Case Laws, Amendments, Statutes, Gazettes, and Agreements [8]. Thus, a proper system is required for the management of this information. As the country adopted digitization in the recent past, there is a large void in the legal information system domain. Thus, the country's legal industry suffers from: inconsistencies of legal information, improper management of historic case laws, and unavailability of some important historic legal documents. This has become very problematic to all the stakeholders that have been mentioned above, as the legal knowledge of a country should be highly accurate, efficient, and effective for the proper enforcement of the rule of law in a country. This could also be understood by a study conducted by A. Cammelli and E. Fameli where the need for a proper legal information retrieval system that could handle the various legal information types has been highlighted [9].

1.2 Research Gap

In considering recent advancements in modern technology, individuals have initiated an exploration for resolutions to issues arising from the utilization of modern technology and systems. This trend has revealed itself within the legal sector as well. The increasing amount of information within the legal sector has become a significant phenomenon, prompting individuals to develop diverse methods aimed at addressing these challenges.

There have been notable advancements in the area of legal information retrieval, indicating a substantial body of research conducted globally in the field of legal information retrieval.

The syntaxes and semantics of legal information and documents vary across different countries worldwide. The variation in cultural, traditional, geopolitical, and international influences experienced by different countries and their legal systems leads to differences in the concepts, vocabulary, and structure of legal information across jurisdictions. The primary research need observed is the lack of a legal information retrieval system specifically designed to address the unique requirements of the Sri Lankan legal landscape. Based on the conducted literature survey, it was determined that the systems developed primarily prioritize the legal systems specific to the corresponding country. Based on an extensive review of existing literature and a comprehensive analysis of existing systems, it is evident that there are significant gaps that can be addressed through the construction of a Legal Information Retrieval System, as emphasized in this research.

1.2.1 Research carried out in different countries to suit their unique context

- **Research A** - This [10] research paper which was written by Diana Rudenko, Kirill Skorniakov, and Yaroslav Nedumov centers on the extraction of legal reference links from Russian legal documents. Its primary objective is to structure the legal references found within these documents. The study exclusively concentrates on isolating the legal reference links, disregarding the remaining content. It is important to note that this research pertains specifically to the Russian language and Russian law.

Two significant research gaps emerge from this study. Firstly, there is a need to expand the developed system's scope to encompass the entire document and extract key information comprehensively, rather than solely focusing on legal references. Secondly, there is a distinct gap in the creation of a tailored system designed to cater to the nuances of the Sri Lankan legal domain. Addressing these

gaps would not only enhance the efficiency of legal document analysis but also facilitate broader applications across diverse legal contexts.

- **Research B** - Another notable research endeavor pertains to the development of an automated legal document summarization system in India [11]. A significant void was identified during this investigation, notably the absence of a similar system within the Sri Lankan legal domain. Sri Lanka's legal landscape presents a unique context, where numerous legal cases exhibit interconnections. This is evident when examining the Supreme Court of Sri Lanka's website [12], which houses extensive judgments for these cases. The voluminous nature of these documents poses a substantial challenge for legal professionals and law students seeking to grasp case details, demanding a considerable investment of time. Consequently, there is a pressing demand for a comparable summarization system tailored to the local legal context.
- **Research C** - Another noteworthy research undertaking, the JudgeDoll system [13], focused on summarizing supreme court judgments in Thailand. A critical research gap emerged from this study: the absence of a parallel system designed for Sri Lanka's legal domain. This gap underscores the need for an indigenous legal document summarization tool to cater to the specific requirements of Sri Lanka's legal landscape.

Additionally, a research initiative conducted in India [14] explored legal case document summarization using an extractive approach, a method that has not been currently implemented within the Sri Lankan legal domain. This highlights yet another opportunity for innovation and the development of a customized summarization system to serve the legal community in Sri Lanka effectively.

1.2.2 Problems in low accuracy similarity calculations

In the process of information retrieval (IR), the calculation of similarities plays a pivotal role in extracting the most relevant information. The accuracy of the methodology employed for similarity calculation is, therefore, a fundamental determinant of the

effectiveness of the proposed system. Upon conducting a comprehensive literature survey, it became evident that similar systems to the one we have developed have often relied on older and less accurate techniques for similarity calculation.

- **Research D** - Conducted by Huu-Thanh Duong and Bao-Quoc Ho, who proposed a QnA (Question and Answer) system tailored for Vietnamese legal documents. Their approach involved selecting and extracting answers from pertinent documents using a similarity calculation technique, which yielded an experimentally achieved precision rate of nearly 70% [15]. However, it's essential to note that their answer selection method relied on calculating similarity scores using tf-idf (Term Frequency-Inverse Document Frequency). This approach, while effective to some extent, lacks the capability to discern contextual relationships between words. Consequently, this method left room for improvement within the research context, and it highlighted an opportunity for the developed system to enhance the accuracy of its models by employing more advanced and refined methodologies.

The criticality of accuracy in legal decision support systems cannot be overstated, as the entire domain relies heavily on the precision of the information it consumes. Recognizing this need, the developed system was specifically designed to bridge this research gap by elevating the accuracy of the retrieved documents and information. To achieve this, the system leverages transformer-based language models, which represent a cutting-edge advancement in the field of information retrieval and natural language processing [16].

1.2.3 The problem of low performance in pre-trained language models

In the realm of Natural Language Processing (NLP), one of the latest and most groundbreaking methodologies involves the application of pre-trained language models, specifically those based on transformer architecture. Through extensive exploration of this domain, it became apparent that these advanced technologies could be harnessed to enhance the performance of the developed system. However, it's essential to recognize that these models, although powerful, are not without their unique challenges.

- **Research E** – Delves into Bidirectional Encoder Representations from Transformers (BERT). BERT, developed by Kenton, Jacob Devlin Ming-Wei Chang, and Lee Kristina Toutanova at Google AI, stands out as a prominent transformer-based language model [17]. BERT is pre-trained on a vast and diverse dataset, making it a versatile and widely adopted tool in the field of decision support systems. Nevertheless, it's crucial to note that, despite its utility in various use cases, BERT's generalized nature limits its direct applicability to a Sri Lankan legal knowledge base. This limitation underscores an opportunity for improvement, particularly in tailoring these transformer-based models to the specific requirements of the Sri Lankan legal context. The key innovation lies in adapting and fine-tuning these pre-trained models to cater to the nuances of Sri Lankan legal language, terminologies, and intricacies. By doing so, it becomes possible to significantly enhance their effectiveness and relevance within the Sri Lankan legal domain. This adaptation process is essential to bridge the gap between the powerful capabilities of pre-trained language models like BERT and the unique characteristics of the local legal landscape.

In conclusion, it could clearly be understood that several research gaps were essentially addressed by the developed system. The developed system was planned and constructed to provide solutions to these research gaps.

Table 1 Research gap representation

	Research A	Research B	Research C	Research D	Research E	Built System
Supports Sri Lankan legal framework	✗	✗	✗	✗	✗	✓
Document summarization	✗	✓	✗	✓	✓	✓
Domain-specific knowledge	✓	✓	✗	✗	✗	✓
High Accuracy	✗	✗	✗	✗	✗	✓

1.3 Existing Systems and Feature Comparisons

In the current landscape, legal professionals predominantly rely on two systems:

1. LawLanka (<https://www.lawlanka.com>)
2. LawNet (<https://www.lawnet.gov.lk>)

Both platforms offer valuable legal resources, including Sri Lankan law reports, Supreme Court judgments, and Court of Appeal judgments, as well as a repository of acts and codes. However, these platforms differ significantly in terms of features and accessibility.

- **System 1:** LawLanka, a robust resource, provides advanced features such as keyword-based document searches. Making it a powerful tool for legal practitioners seeking specific information. However, it comes with a significant drawback. It operates on a paid subscription model. Users are required to pay for access, which can be a financial barrier for law students and individuals with limited resources.

- **System 2:** LawNet, offers a free service, making it accessible to a broader audience, including law students. However, it has limitations in terms of features. The platform primarily provides documents in their raw format, typically in PDFs, without advanced search capabilities or structured document formats.

The newly developed “Sri Lankan Legal Information Retrieval System” represents a noteworthy advancement in addressing these challenges. It offers a comprehensive solution that combines accessibility with functionality.

The system introduces structured document formats, enabling users to navigate legal documents with ease. The implementation of a keyword search feature further enhances the user experience, facilitating quick and precise information retrieval. Perhaps most significantly, the system excels in extracting relevant information from documents, sparing users the arduous task of sifting through entire texts. Furthermore, the system is powered with legal document summarization and legal decision support functionalities. It empowers legal professionals, law students, and the wider community with a user-friendly and cost-effective solution, ultimately democratizing access to legal knowledge and streamlining the way legal information is harnessed and utilized.

Table 2 System comparison

	System 1	System 2	Built System
Financially affordable to a wide range of users	✗	✓	✓
Document summarization	✗	✗	✓
Structured document nature	✗	✗	✓
Decision support	✗	✗	✓
User-friendly environment	✗	✗	✓

2 RESEARCH PROBLEM

The increasing number of legal papers in Sri Lanka poses a complex and varied challenge within the legal domain. The matter at hand contains various interrelated facets, all of which contribute to the intricate nature and importance of the issue.

Within the legal framework of Sri Lanka, various entities assume crucial functions, encompassing government offices, courts, law firms, and corporations. The precise identification and extraction of these items are crucial for improving search accuracy and enabling thorough comprehension. [18]

The inclusion of a Legal Document Summarization function is of utmost importance in the advancement of a legal information retrieval system in Sri Lanka. The utilization of automatic summarizing can greatly improve the usability and accessibility of legal material, considering the typically extensive and intricate character of legal documents.

In Sri Lanka, the abundance of legal documents such as judgments, legislation, and regulations poses a challenge for legal practitioners and the general populace in effectively extracting pertinent information. The methodologies used for summarizing documents should be customized in order to effectively capture the essential elements of the content while preserving the important legal principles, precedent cases, and legal reasoning.

Another issue that arises is the absence of legal knowledge hubs in existing systems. Legal ontologies play a fundamental role in comprehending the semantic framework of legal data, hence facilitating accurate retrieval of legal information in the context of Sri Lanka. The development of comprehensive legal ontologies that are specifically designed for the legal system of Sri Lanka is a substantial research challenge.

A survey was undertaken among members of the legal community, specifically legal professionals such as law students and lawyers, to identify the unique issues they face in their day-to-day professional activities. The survey results have provided significant insights into the urgent needs within the field of law. The participants emphasized the importance of a comprehensive legal information retrieval system that would facilitate

their access to legal materials. The implementation of this technology would significantly augment the efficacy of legal research and the process of case preparation.

The figure provided below visually depicts the urgent need for a legal information retrieval system. This statement highlights the intricate nature and difficulties involved in dealing with legal papers, emphasizing the significance of establishing a structured framework to improve the retrieval of documents and accessibility of data within the legal field.

Do you have a need for a proper, comprehensive and accurate legal information system?

175 responses

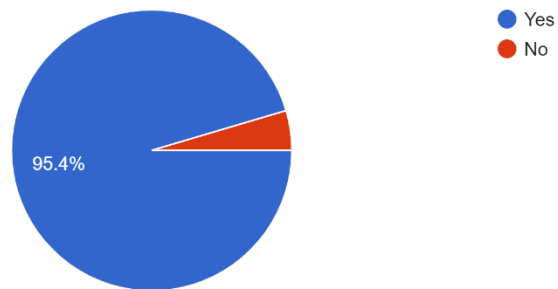


Figure 2 Suggested requirements by the local legal professionals.

3 OBJECTIVES

3.1 Main Objective

The implemented legal information retrieval system, AYCA aims to provide a user-friendly and multifaceted solution for retrieving legal information in the Sri Lankan legal context that is efficient, accurate, and consistent.

The implemented system has made it easier for legal experts to make informed decisions by analyzing legal case reports and providing decision support. It has been a valuable resource for both lawyers and laypeople in Sri Lanka seeking legal information. Furthermore, the system has allowed users to locate all relevant information related to the case or information sought in legal case reports.

The implemented system has fulfilled the main objective and has addressed the lack of legal information by providing consistent access to such documents, thereby revolutionizing Sri Lanka's legal industry. Its time-saving legal information search capabilities have benefited both lawyers and law students, resulting in higher legal industry standards in the country.

3.2 Specific Objectives

3.2.1 Data extraction and structuring.

This specific goal of this study is to tackle the challenges posed by disorganized legal documents in Sri Lanka. It does this by creating a structured framework within a larger system called "AYCA," which is a Legal Information Retrieval System in Sri Lanka. This framework aims to make it easier for legal professionals to find important information in legal documents quickly, saving them time when dealing with large amounts of text. By organizing legal documents and using advanced techniques to find information, this research aims to make legal materials more accessible and useful, ultimately making life easier for legal practitioners.

3.2.2 Legal knowledge graph development.

This research also aims to build a strong legal knowledge graph specifically designed for Sri Lanka. This graph will be focused on Supreme Court Verdicts and will serve as the foundation for a comprehensive legal information retrieval system. By implementing this knowledge graph, it will become much easier for legal professionals, including lawyers, practitioners, and students, to search, retrieve, and review legal case reports and related documents within Sri Lanka's legal system. They will have advanced search options, such as filtering by case numbers, parties involved, dates, judges, and legal citations. Ultimately, this will make legal information more relevant and accessible.

3.2.3 Legal document summarization.

This section explains the specific goal, the legal document summarizer. The goal is to help lawyers, law students, and the general public, who were identified as key users, quickly understand lengthy legal case documents. To achieve this, the summarizer must create very short yet accurate summaries that capture the most important parts of the document. To ensure accuracy and relevance, these summaries need to be validated by a legal expert because even if the model is accurate, it might miss crucial details. So, the main objective here is to provide users with brief, accurate summaries that include the essential information they need from the document, as requested.

3.2.4 Legal question and answering.

Another sub-objective of this research is to create a sophisticated legal decision support system customized for Sri Lankan laws. This system centers on an advanced question-answering mechanism, aiming to provide accurate legal answers to users, including legal professionals and students. It uses cutting-edge technology, specifically a transformer-based approach, which uses advanced natural language processing and machine learning to understand complex legal language and deliver precise responses. This research has a dual purpose: bridging the gap between Sri Lankan legal documents and accessible legal knowledge and simplifying legal research to provide quicker and more reliable answers to legal questions.

4 METHODOLOGY

The methodology employed in this research project has been meticulously designed to tackle the goal of implementing a systematic legal information retrieval system for Sri Lankan Supreme Court judgments and laws, with the aim of enhancing the accurate retrieval of legal information. The Sri Lankan Legal Information Retrieval System serves as a comprehensive knowledge base within the Sri Lankan legal framework, specifically emphasizing Supreme Court verdicts and the laws pertaining to the Narcotic Drug Act. The present methodology is organized into distinct stages, each of which assumes a critical function in the advancement of the system and the retrieval of relevant case documents, case summaries, and responses to user questions. During the process of this research, the methodology employed adhered to an agile development approach, which effectively enables flexibility and gradual improvements. In the Sri Lankan legal domain, the utilization of a user-friendly web application interface that incorporates the knowledge graph holds the potential to enhance the experience of end users. This interface would facilitate efficient and accurate searching and retrieval of case information. Additionally, the integration of the BART model for case summary generation and the RoBERTa model for question answering further enhances the functionality of the application. Ultimately, these advancements aim to benefit end users by providing them with a more streamlined and effective means of accessing and utilizing legal information.

4.1 Requirement Gathering and Analysis

The requirement gathering and analysis phase is a crucial aspect of the software development life cycle, without which the cycle would be considered incomplete. Stakeholder evaluation is the process of finding, documenting, and understanding the interests of those who will be affected by the software solution. During the process of requirement collection, the development team engages in communication with the stakeholders of the project in order to gather and document the needs expressed by these stakeholders. The criteria should be clearly specified and capable of being observed. After

the collection of requirements, a thorough evaluation is conducted to evaluate their feasibility, alignment with the project's objectives, and potential to provide the anticipated outcomes. Ensuring the completion of this step is crucial in order to mitigate the potential financial consequences associated with future changes or revisions during the developmental phase. By adhering to this practice, the project maintained its projected timeline and effectively mitigated any issues pertaining to scope. Furthermore, the team was able to mitigate potential risks and address issues by undertaking comprehensive requirement gathering and analysis throughout the first stages.

The first stage of the project involved the process of obtaining requirements. During this phase, it is essential to accurately define and document all user needs clearly and unambiguously. Utilizing a Software Requirement Specification (SRS) document proved advantageous in this context.

Initially, business issues were identified through the utilization of a questionnaire and the conducting of interviews with end users, specifically law school students. The objective of the study was to investigate public opinion regarding the proposed product, specifically examining whether good views are prevalent in comparison to negative opinions. The questionnaire was distributed to a diverse group of participants, including lawyers, legal professionals, law school students, and the public. The survey was conducted using a Google Form. The process of conducting interviews involved the utilization of telephone calls and meetings in person. During this phase of the interview process, the interviewees engaged in a discussion regarding the issues associated with the current systems in place. Additionally, they explored how the proposed system might offer resolutions to these identified problems. The questionnaire examined the level of public approval for the planned paid subscription plan. Furthermore, as a part of the research plan, it was necessary to identify limitations and shortcomings present in the current systems. Please refer to the [APPENDIX A](#) for the questionnaire.

4.1.1 Functional Requirements

- Able to ensure the precise extraction of legal entities from the Supreme Court judgments accessible on the official website of the Sri Lankan Ministry of Justice's Supreme Court.
- Able to construct a knowledge graph as the knowledgebase to develop complex relationships among identified legal entities from many cases, and therefore enhance the effectiveness of information retrieval and navigation.
- Able to generate and provide a concise, insightful summary for an inserted PDF document of a legal case.
- Able to provide reliable answers to user questions based on the Narcotic Drug Act of Sri Lanka.
- The platform should provide a user-friendly interface that facilitates easy access and interaction for legal professionals and students.
- Users should be able to search and retrieve legal information efficiently using relevant searching criteria.

4.1.2 Non-Functional Requirements

- To ensure the credibility of the information, the system must maintain high levels of accuracy in legal entity extraction, summarization, and question answering.
- To boost user productivity, the system must have a quick response time and efficient information retrieval.
- The platform should be scalable to accommodate an expanding knowledge graph to represent the knowledge base.
- The protection of sensitive legal information is of the highest priority. (this information is publicly available, so it doesn't cause any problem with confidentiality)
- The user interface must be intuitive, with straightforward navigation and accessibility for both legal professionals and students.

4.2 Feasibility Study

- **Schedule Feasibility:** The assessment of the proposed system's schedule feasibility is an essential factor in determining its overall viability. The addition of a project timeline, consisting of clearly defined milestones and deliverables, served to ensure that the development and implementation processes can be accomplished within the designated timeframe. The feasibility element highlighted the system's practicality by indicating its ability to be implemented promptly to tackle the highlighted issues in legal information retrieval.
- **Technical Feasibility:** The primary emphasis lies in identifying the essential technological prerequisites for system development. Team members were expected to possess a comprehensive understanding of programming languages such as Python and similar counterparts, along with proficiency in diverse development technologies. Expertise in machine learning algorithms, natural language processing algorithms, and data extraction and processing formed a critical requirement. Furthermore, an in-depth comprehension of software development methodologies and associated tools and technologies was deemed essential for the project's success.
- **Economic Feasibility:** The developed system demonstrates economic viability across various aspects. The utilization of cost-effective, open-source models such as BART and RoBERTa led to a reduction in development expenditure. Some expenditures had to be spent to find learning resources.
- **Legal Feasibility:** The system has been developed with careful consideration of both legal and ethical aspects. The system functions within the constraints of copyright legislation, thereby guaranteeing the ethical use of lawful documents. Furthermore, strict compliance with privacy and data protection requirements is maintained when handling confidential legal information.

4.3 System Design

4.3.1 High-level overall system architecture diagram

The Sri Lankan legal information retrieval system is made up of four main parts: the data extraction and structuring component, the development of a legal knowledge graph and retrieval of legal documents component, the summarization of legal documents component, and the legal question-answering component.

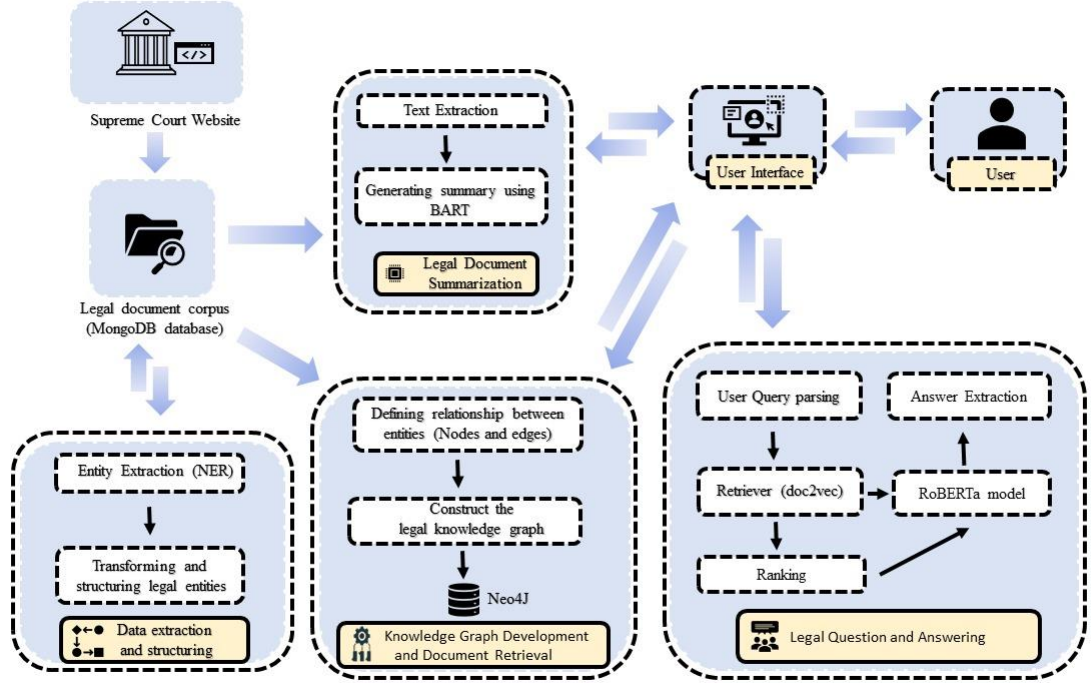


Figure 3 System Architecture Diagram

4.3.1.1 Data Extraction & Structuring Module

The first stage of the system is carried out by the Data Extraction and Structuring component, which is responsible for automatically extracting information and organizing Supreme Court Judgments. This component serves to enhance the process of accessing legal documents in PDF format from the official website of the Supreme Court of Sri Lanka. The study employs a specialized Named Entity Recognition (NER) model that has been trained exclusively for the analysis of Supreme Court decisions in Sri Lanka. This model is applied to identify and extract significant legal entities, including case titles, acts,

legal citations, counsel, judges, argued dates, verdict dates, judgments, and other important entities [19]. The entities that have been extracted are methodically arranged and organized into a MongoDB database.

This component guarantees the presence of a thorough collection of meticulously organized legal documents, serving as the data repository for the remaining components inside the system.

4.3.1.2 Legal Knowledge Graph Development and Document Retrieval Module

The incorporation of the Legal Knowledge Graph Development and Document Retrieval component is a vital component in enhancing the legal information retrieval capabilities of the system [20]. This component greatly improves the ability to obtain legal case knowledge by utilizing a process that involves the identification of relationships between legal entities, that have been extracted by the data extraction and structuring component of the system, and the construction of a knowledge graph to represent the underlying knowledge base.

By using technologies such as Neo4j and the Cypher query language, the system is capable of producing a well-organized and interconnected network of legal data, which in turn enables seamless navigation and comprehensive analysis [21]. The utilization of a knowledge graph offers significant advantages to those in the legal field, including attorneys and law students. It serves as a user-friendly tool that facilitates the retrieval of extensive case information, therefore enhancing the efficiency and effectiveness of legal research [22].

4.3.1.3 Legal Document Summarization Module

The Legal Document Summarization component of the system is designed to produce concise and insightful abstract summaries of Supreme Court Verdicts. The abstractive summarization techniques employed by the model involve the rephrasing and compressing of the source material to generate summaries using its own language. The procedure of data collection and preprocessing involves selecting Supreme Court case files from the official website, namely those from the period spanning 2015 to 2020.

The preprocessing step involves the removal of stop words and unnecessary data in order to clean and organize datasets [23]. The Bidirectional-Auto-Regressive-Transformer (BART) model is subsequently trained using preprocessed Supreme Court Verdicts of Sri Lanka. The BART model is employed for abstractive summarization, which involves the rephrasing and compressing of case file data into concise and useful summaries [24]. This characteristic enhances the comprehensibility and easy understandability of legal case material for legal professionals, students, and anyone in search of legal knowledge [25].

4.3.1.4 Legal Question & Answering Module

The Question and Answer (Q&A) component is a valuable addition to the system, providing users with numerous advantages. It enables efficient and precise access to information regarding Sri Lanka's Narcotic Drugs and Psychotropic Substances Control Act, making legal research easier [26]. This feature allows users, including attorneys, researchers, and the public, to rapidly obtain accurate answers to their questions regarding Sri Lankan narcotics law. This streamlined access to legal information increases productivity, facilitates the making of informed decisions, and facilitates legal analysis. The Q&A component significantly reduces the time and effort required for legal research, making it an indispensable resource for those seeking thorough and trustworthy insights into this area of the law [27].

4.4 Understanding the Key Pillars

The development and implementation of this Sri Lankan legal information retrieval system, centered on Supreme Court Verdicts, are dependent on several key pillars that form the basis of this research project. These pillars collectively contribute to the construction of a robust and user-friendly system tailored to the unique requirements of the legal domain in Sri Lanka. The following pillars are at the core of this component:

4.4.1 Natural language processing

The study is dependent on Natural Language Processing (NLP) as a fundamental element throughout all its components, encompassing the BART summarization model, RoBERTa

question-answering model, and Spacy NER for information extraction. Natural Language Processing (NLP) facilitates the comprehension and interpretation of human language by machines, hence assisting the legal information retrieval system. This technology plays a crucial role in extracting valuable insights from real-world data inputs. The NLP (Natural Language Processing) process includes various stages, such as data pre-processing and algorithm development, as well as the cleaning and organization of datasets. Additionally, it involves the extraction of crucial information from intricate legal documents. Natural Language Processing (NLP) serves as the foundation for the four fundamental elements of the system, enabling the operational capabilities of summarization, question-answering, and information extraction models.

4.4.2 Named entity recognition

Named Entity Recognition (NER) plays a fundamental role in the field of Natural Language Processing (NLP). NER serves as a language detective, proficient in the identification and categorization of elements inside unorganized textual input. The study examines an extensive collection of Sri Lankan Supreme Court rulings, which include a wide range of legal entities including case numbers, legal citations, judges, judgments, parties, and other relevant information. Named Entity Recognition (NER) extends its scope beyond simple recognition by implementing a systematic categorization of these crucial entities. The utilization of this methodical technique plays a pivotal role in the construction of a knowledge graph that establishes connections among legal case data, so NER forms an essential key pillar of the system's methodology [10].

4.4.3 Knowledge graphs

A knowledge graph is an advanced representation for representing data that is designed to capture entities and complex relationships within an area of study. Knowledge graphs consist of two primary components, namely nodes and edges. Within the area of legal cases, there exists a collection of nodes that represent various legal entities, including case numbers, judges, dates, parties involved, and legal concepts. Additionally, some edges indicate the connections and relationships between these entities. The presented system

effectively illustrates the interrelationships observed in Supreme Court rulings, hence improving understanding, and streamlining the retrieval of legal information. The graph's semantic context, network structure, and scalability provide valuable tools that empower users to effectively investigate, retrieve, and analyze detailed legal case data. Consequently, it serves as a robust knowledge repository for legal information retrieval within the scope of this research project [28].

4.4.4 Big data technologies

One of the core elements of data science is the creation of the knowledge graph, which serves as a significant foundation of big data technologies. Neo4j, an elastic graph database technology, functions as the foundational structure for the organization and management of the extracted legal case entities and the relationships between them. The advent of this knowledge graph represents the advancement of a comprehensive collection of legal information, facilitating efficient search and retrieval of data inside an organized structure utilizing graphs [29].

4.5 Methodology

4.5.1 Definition of system scope and identification of data sources

For the developed system, the scope definition and data sources were meticulously determined through a process of careful consideration and expert consultation. After a thorough evaluation, it was decided to focus on Supreme Court verdicts for the year 2015 and the Conventions Against Illicit Traffic in Narcotic Drugs and Psychotropic Substances Act, No. 1 Of 2008, during the narrowing down of the scope to create the system.

The following reputable sources were identified as the primary data sources for information extraction, ensuring comprehensive coverage and reliability:

- Ministry of Justice (www.lawnet.gov.lk)
- Court of Appeal of Sri Lanka (courtofappeal.lk)
- Supreme Court of Sri Lanka (supremecourt.lk)

- Department of Government Printing (documents.gov.lk)

It is noteworthy that all these sources are open source, aligning with the project's commitment to ethical and legal data collection practices. This approach not only upholds transparency and accessibility of legal information but also maintains compliance with ethical standards, ultimately contributing to the system's credibility and reliability.



The screenshot shows the homepage of the Supreme Court of Sri Lanka. The header is red with the court's emblem and name. Below the header is a navigation bar with links like HOME, ABOUT THE COURT, HISTORY, E-FILING, DAILY COURT LIST, BENCH LISTS, JUDGEMENTS, SETTINGS, INFORMATION TO LITIGANTS, SPECIAL/URGENT NOTICES, and TEST. The main content area is titled 'Judgments Delivered in 2015' and contains a table with columns for DATE, CASE NUMBER, NAME OF PARTIES, COURT NUMBER, and FILE. The table lists several judgments, including one from 15/12/15 (SC Appeal No. 15/2013) involving Jathika Sevaka Sangamaya and another from 14/12/15 (SC/FR 1006/2009) involving Hapugodage Jagath Perera.

DATE	CASE NUMBER	NAME OF PARTIES	COURT NUMBER	FILE
15/12/15	SC Appeal No. 15/2013	Jathika Sevaka Sangamaya, No. 416, Kotte Road, Pitakotte. (On behalf of S.S. Samarasinghe, G.V.A.N. Senadheera, S.N. Nanayakkara, P.B.H. Denuwara and Kalyani Samarakoon) Applicant Vs. Sri Lanka Hadabima Authority No. 08, Gannoruwa Road, Peradeniya. Respondent AND Jathika Sevaka Sangamaya No. 416, Kotte Road, Pitakotte (On behalf of S.S. Samarasinghe, G.V.A.N. Senadheera, S.N. Nanayakkara, P.B.H. Denuwara and Kalyani Samarakoon) Applicant-Appellant Vs. Sri Lanka Hadabima Authority No. 08, Gannoruwa Road, Peradeniya. Respondent-Respondent AND NOW BETWEEN Jathika Sevaka Sangamaya No. 416, Kotte Road, Pitakotte (On behalf of S.S. Samarasinghe, G.V.A.N. Senadheera, S.N. Nanayakkara, P.B.H. Denuwara and Kalyani Samarakoon) Applicant-Appellant-Appellant Vs. Sri Lanka Hadabima Authority No. 08, Gannoruwa Road, Peradeniya. Respondent-Respondent-Respondent		DOC
14/12/15	SC/FR 1006/2009	Hapugodage Jagath Perera Petitioner Vs 1. Gothami Ranasinghe Inspector of Police, Officer-in-Charge of the Minor Crime Branch, Police Station, Mirigama. 2. Mila Vitharana alias Milavithanachchi Inspector of Police, Officer-in-Charge of the Traffic Branch, Police Station, Mirigama. 3. Milinda Premaratne Karunaratne, Sub Inspector of Police, Police Station, Mirigama. 4. Inspector General of Police, Police Head Quarters, Colombo. 5. Hon. Attorney-General Attorney General's Department Colombo 12. Respondents		DOC
13/12/15	S.C. Appeal No. 44/2012	Padmal Anyasin Mendis, No.29, Moratumulla Road South, Moratuwa. Plaintiff Vs. Vijith Abraham de Silva, No. 13, Peduru Mawatha, Moratumulla, Moratuwa. And Vijith Abraham de Silva, No. 13, Peduru Mawatha, Moratumulla, Moratuwa Defendant-Appellant Vs. Padmal Anyasin Mendis, No.29, Moratumulla Road South, Moratuwa. Plaintiff-Respondent And Now Between Vijith Abraham de Silva, No. 13, Peduru Mawatha, Moratumulla, Moratuwa Defendant-Appellant- Petitioner-Appellant Vs. Padmal Anyasin Mendis, (Deceased) No.8108, Halgahadeniya Road, Gothatuwa. Plaintiff - Respondent Respondent-Respondent Sarukali Patabedige Claris de Silva, of No. 6108, Halgahadeniya Road, Gothatuwa. Substituted Plaintiff-Respondent- Respondent- Respondent		DOC
09/12/15	SC Appeal 27/2013	Halgahadeniya Thilakarane (Deceased) Plaintiff RAG Sumanawathi Substituted Plaintiff Vs Galkaduwegedara Sunil Jayathilake Defendant AND BETWEEN RAG Sumanawathi Substituted-Plaintiff-Appellant Vs Galkaduwegedara Sunil Jayathilake Defendant-Respondent AND NOW BETWEEN RAG Sumanawathi Substituted-Plaintiff-Appellant Vs Galkaduwegedara Sunil Jayathilake Defendant-Respondent		DOC

Figure 4 Source website

4.5.2 Data acquisition and preparation

In the context of the system, data acquisition and preparation form the crucial foundation for effective information retrieval. This phase involves meticulous sourcing and structuring of legal documents from trusted authorities. By adhering to rigorous standards and ethical practices, the system ensures the integrity and accessibility of legal information.

4.5.2.1 Data preparation for the Data Extraction & Structuring Module

Following the completion of the data collection phase, the gathered files underwent a preliminary preprocessing step. Given the sensitive nature inherent to legal documents, it became important to maintain the highest levels of accuracy throughout the annotation process. Consequently, the chosen methodology for identifying and extracting specific named entities from each document was manual annotation. These named entities

encompassed various elements such as case numbers, court names, plaintiffs, defendants, argued dates, decided dates, legal counsel, judges, and individual judgments.

To facilitate this manual annotation process effectively, an online annotation tool known as NER Annotator was utilized, accessible at <https://tecoholic.github.io/ner-annotator/>. This tool offered a user-friendly interface that enabled annotators to meticulously pinpoint and label named entities within the textual content of each file. Employing this tool ensured that the identification of entities was consistent and precise, aligning seamlessly with the accuracy standards mandated by the sensitive legal data under scrutiny.



Figure 5 Online NER annotation tool

The manual annotation process culminated in the creation of structured JSON files. These JSON files not only contained the cataloged named entities but also provided detailed annotations specifying their precise locations within the context of the entire document. This meticulous level of annotation served as invaluable training data for the subsequent development of a customized Named Entity Recognition (NER) model.

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            985,
            986,
            "LEGAL_CASE_REFERENCE"
          ],
          [
            987,
            988,
            "LEGAL_CONCEPT"
          ],
          [
            989,
            990,
            "COURT"
          ],
          [
            991,
            992,
            "COUNSEL"
          ],
          [
            993,
            994,
            "JUDGE_1"
          ],
          [
            995,
            996,
            "JUDGE_2"
          ],
          [
            997,
            998,
            "JUDGE_3"
          ],
          [
            999,
            1000,
            "JUDGEMENT_1"
          ],
          [
            1001,
            1002,
            "JUDGEMENT_2"
          ],
          [
            1003,
            1004,
            "JUDGEMENT_3"
          ],
          [
            1005,
            1006,
            "DEFENDENT"
          ],
          [
            1007,
            1008,
            "PLAINTIFF"
          ],
          [
            1009,
            1010,
            "OTHER_PARTIES"
          ],
          [
            1011,
            1012,
            "CASE_NO"
          ],
          [
            1013,
            1014,
            "LEGAL_CASE_REFERENCE"
          ],
          [
            1015,
            1016,
            "LEGAL_CONCEPT"
          ],
          [
            1017,
            1018,
            "COURT"
          ],
          [
            1019,
            1020,
            "COUNSEL"
          ],
          [
            1021,
            1022,
            "JUDGE_1"
          ],
          [
            1023,
            1024,
            "JUDGE_2"
          ],
          [
            1025,
            1026,
            "JUDGE_3"
          ],
          [
            1027,
            1028,
            "JUDGEMENT_1"
          ],
          [
            1029,
            1030,
            "JUDGEMENT_2"
          ],
          [
            1031,
            1032,
            "JUDGEMENT_3"
          ],
          [
            1033,
            1034,
            "DEFENDENT"
          ],
          [
            1035,
            1036,
            "PLAINTIFF"
          ],
          [
            1037,
            1038,
            "OTHER_PARTIES"
          ],
          [
            1039,
            1040,
            "CASE_NO"
          ],
          [
            1041,
            1042,
            "LEGAL_CASE_REFERENCE"
          ],
          [
            1043,
            1044,
            "LEGAL
```

4.5.2.2 Data preparation for the Legal Knowledge Graph Development and Document Retrieval Module

For this component, a robust data pipeline has been created to facilitate the development of the Legal Knowledge Graph (LKG) and document retrieval module. This pipeline effectively bridges the gap between MongoDB, where legal entities have been meticulously stored through the employment of Named Entity Recognition (NER), and Neo4j, where the LKG will be constructed. The data transformation and integration process are central to this pipeline. It involves converting raw legal entity data, organized into a graph structure that represents all the interconnections and relationships within multiple cases. This transformation is optimized for LKG integration while ensuring that the LKG captures comprehensive legal relationships. This includes establishing meaningful connections between legal entities through entity linking and constructing various relationships within the knowledge graph, such as “Who judged a case?”, “whom are the parties involved in a case?”, “When was the case argued?”, “Who are the counsel?”, “What are the legal references cited in a case?”, and furthermore.

A significant part of this process involves enriching the dataset by adding extra details like attribute information to each relationship while creating them. This helps us better understand legal entities and connections within multiple cases. The pipeline works in an organized way, smoothly moving data from MongoDB to Neo4j while transforming. Each step is clearly defined, focusing on processing and transforming data. This approach uses a specific custom Python script for these tasks, and it also makes sure the pipeline can handle lots of data efficiently.

After creating the above-mentioned pipeline, detailed transformation steps, and legal knowledge graph creation steps are discussed in the Implementation section.

4.5.2.3 Data preparation for the Legal Document Summarization Module

The dataset used for this module is structured into three columns: Case ID, Context, and Summary, with each row representing a single case file. Notably, the Case ID column, deemed unnecessary for generating judgment summaries, was omitted. It's worth noting

that the dataset received approval from the external supervisor (refer to [APPENDIX D](#)), affirming its relevance to the research.

Tokenization, a critical step in Natural Language Processing (NLP), was employed in this research, particularly due to two key reasons:

- Deep learning models like BART require fixed-size inputs, making tokenization vital in converting variable-size inputs into a consistent format.
- Tokenization aids in preprocessing, and standardizing input text by eliminating unnecessary whitespaces, ensuring uniformity.

The BART tokenizer, initialized with a pre-trained model "facebook/bart-large," was utilized for tokenization. This model had undergone extensive pre-training on a diverse corpus of text and fine-tuning for various NLP tasks, including text summarization. The "encode_plus" function within the BART tokenizer was employed for tokenizing the legal dataset while incorporating other essential tasks. Notably, the "Context" section, containing lengthy judgments, was tokenized with a maximum sequence length of 512 tokens, while the "Summary" section had a maximum sequence length of 150 tokens.

By following these meticulous steps, the dataset was effectively prepared for both the training and evaluation phases. The tokenization process's significance lies in its role in ensuring model compatibility and facilitating the uniform treatment of input data. The accompanying image provides a visual snapshot of the "encode_plus" tokenization function. This comprehensive approach to tokenization serves as a foundational step in the development of a robust BART-based legal document summarization model.

```
tokenizer = BartTokenizer.from_pretrained("facebook/bart-large", add_prefix_space=True)

def encode_data(tokenizer, context, summary):
    inputs = tokenizer.encode_plus(context, add_prefix_space=True, padding='max_length', truncation=True, max_length=512, return_tensors='pt')
    labels = tokenizer.encode_plus(summary, add_prefix_space=True, padding='max_length', truncation=True, max_length=150, return_tensors='pt')
    return inputs, labels
```

Figure 7 Tokenization function - summarization model

4.5.2.4 Data preparation for the Legal Question & Answering Module

Manual annotation of data:

To uphold the highest standards of dataset quality and accuracy, a meticulous manual annotation process was employed. Utilizing a dedicated annotation tool, illustrated in the below diagram. This comprehensive procedure involved the intricate manual alignment of legal queries with their corresponding answers extracted from the legal documents. This rigorous approach played a pivotal role in capturing the intricate nuances of the legal context and critical insights.



Figure 8 Manual annotation of documents

SQuAD format creation:

Following the manual annotation process, the system obtained CSV (Comma-Separated Values) files, comprising crucial data elements such as legal context, user-generated queries, answer spans, and answer starting indices.

To align with the standards commonly used for training and evaluating question-answering models, particularly the SQuAD (Stanford Question Answering Dataset) format, a Python script was meticulously developed. This script automated the conversion process, effectively restructuring the dataset to align with the SQuAD format. The transition enhanced compatibility with machine learning models designed for question answering and ensured scalability, consistency, efficiency, and dataset reproducibility.

```

1- {
2-   "context": "PARLIAMENT OF THE DEMOCRATIC SOCIALIST REPUBLIC OF SRI LANKA CONVENTIONS
3-   1 OF 2008 [Certified on 23rd January, 2008] Printed on the Order of Government Pub
4-   Republic of Sri Lanka of January 25,2008 PRINTED AT THE DEPARTMENT OF GOVERNMENT P
5-   Psychotropic Substances Act, No. I of 2008 [Certified on 23rd January, 2008] ",
6-   "qas": [
7-     {
8-       "id": "20001QA",
9-       "question": "What legal provisions are established by the 1988 UN Convention?",
10-      "is_impossible": false,
11-      "answers": [
12-        {
13-          "text": "enable Sri Lanka to deal with the international aspects of illicit
14-          mechanism to monitor the import, export and use of the substances set out
15-          "answer_start": 1620
16-        }
17-      ]
18-    }
19-  ]
20- }

```

Figure 9 SQuAD format

4.5.3 Review of prepared datasets and approval

Before finalizing the datasets for the model building and training, a meticulous review and approval process were diligently conducted. Data points extracted, particularly those specific to Sri Lankan law, underwent comprehensive scrutiny and validation. This critical phase ensured that the datasets met stringent standards of accuracy, relevance, and suitability for the research components' objectives.

To reinforce the dataset's integrity and reliability, an industry professional possessing expertise in the legal domain assumed a pivotal role in this process. Their expert evaluation (documented in [APPENDIX D](#)), affirmed the dataset's credibility, aligning it with the highest standards required for robust legal information retrieval and analysis.

4.5.4 Model building and training.

In the subsequent sections, the models have been meticulously constructed and trained using the prepared datasets, tailored to the unique requirements of each component. This approach ensures the models' proficiency and accuracy in delivering precise legal information.

4.5.4.1 Model building and training for the Data Extraction & Structuring Module

During this pivotal stage of the research, the Data Structuring Module assumes a central role within the system. To achieve the predetermined objectives, the following procedural steps were meticulously adhered to. To begin with, the Spacy library was imported, a critical component in the training pipeline due to its inclusion of essential packages. Subsequently, a pre-trained model furnished by Spacy was loaded, which can be customized to align with the specific requirements of the final product.

Following this, the annotated data that was obtained from the preceding phase were put into practical use. Each file is amalgamated to facilitate its input into the model. Consequently, all the necessary training data is meticulously prepared, paving the way for the model's comprehensive training. The model operates by creating a specialized object termed a "document object." Spacy, in the pre-processing phase, tokenizes the text, executes lemmatization, and performs parsing as integral components of the natural language processing (NLP) workflow. After these preparatory stages, hyperparameter tuning was done to maximize the model's performance.

Importing Helper Libraries

```
import spacy
from spacy import displacy
import os
import json
```

Downloading the model in spacy

```
!python -m spacy download en_core_web_lg
```

Figure 10 Importing spacy library.

In this context, the model undergoes training with an increased number of epochs, setting it at 100. The expansion of the epoch count offers a multitude of benefits. It frequently leads to enhanced model performance by allowing the model to fine-tune its parameters and better fit the training data. Moreover, it enables the model to delve deeper into the intricacies of legal texts, capturing nuanced legal entities and relationships necessitating substantial exposure. Legal documents, renowned for their intricate structures and specialized terminology, warrant an extended training duration for the model to aptly grasp these complexities. Consequently, by extending the training timeframe, the model hones its understanding of legal entities and context, culminating in heightened accuracy and an enhanced ability to discern intricate legal references, thereby amplifying its utility in legal document analysis and information retrieval tasks.

```
[training]
dev_corpus = "corpora.dev"
train_corpus = "corpora.train"
seed = ${system.seed}
gpu_allocator = ${system.gpu_allocator}
dropout = 0.1
accumulate_gradient = 1
patience = 1600
max_epochs = 100
max_steps = 20000
```

Figure 11 Epochs increased to 100.

Furthermore, elevating the batch size during training significantly enhances the model's overall performance. In the context of legal documents, characterized by extensive and intricate textual content, larger batch sizes expedite the processing of vast datasets. This empowers the model to learn from a more extensive array of legal entities and contexts in each training iteration. Consequently, not only does this accelerate the training process, but it also promotes enhanced generalization by exposing the model to a diverse spectrum of legal examples. By coupling a larger batch size with the meticulous requirements of the Named Entity Recognition (NER) task within the legal domain, the model becomes more adept at identifying and categorizing intricate legal entities, ultimately elevating its

accuracy and proficiency in extracting crucial legal information from documents. This, in turn, facilitates precise and efficient legal document analysis.

```
[nlp]
lang = "en"
pipeline = ["tok2vec", "ner"]
batch_size = 1000
disabled = []
before_creation = null
after_creation = null
after_pipeline_creation = null
tokenizer = {"@tokenizers": "spacy.Tokenizer.v1"}
```

Figure 12 Increase batch size to 1000.

Through this rigorous and systematic approach, the training of a highly accurate model was achieved, boasting an impressive accuracy rate of 94%. This heightened accuracy not only augments the model's performance during the testing phase but also contributes significantly to the overall enhancement of the system's capabilities and the efficacy of the Data Structuring Module.

```
===== Initializing pipeline =====
✓ Initialized pipeline

===== Training pipeline =====
i Pipeline: ['tok2vec', 'ner']
i Initial learn rate: 0.001
```

E	#	LOSS TOK2VEC	LOSS NER	ENTS_F	ENTS_P	ENTS_R	SCORE
0	0	0.00	2929.99	0.00	0.00	0.00	0.00
5	200	461206.38	62548.72	23.63	32.11	18.69	0.24
10	400	18556.05	18197.62	49.09	47.61	50.66	0.49
15	600	89341.15	13167.68	67.87	71.30	64.75	0.68
20	800	18701.11	12433.37	78.74	79.47	78.03	0.79
25	1000	14581.58	7962.57	75.33	75.58	75.08	0.75
30	1200	12013.06	5094.86	86.48	85.92	87.05	0.86
35	1400	1018.49	2234.72	87.86	87.36	88.36	0.88
40	1600	305.64	1677.11	91.13	91.28	90.98	0.91
45	1800	523.64	1373.29	89.18	89.18	89.18	0.89
50	2000	308.88	1039.91	93.52	93.60	93.44	0.94
55	2200	1183.86	1082.59	92.31	92.16	92.46	0.92
60	2400	878.13	1002.48	94.17	94.40	93.93	0.94
65	2600	137.94	943.61	91.50	93.05	90.00	0.91
70	2800	3583.62	1036.07	93.85	95.12	92.62	0.94
75	3000	300.94	757.11	95.09	94.93	95.25	0.95
80	3200	190.80	599.79	95.41	95.41	95.41	0.95
85	3400	3176.45	738.67	94.91	95.07	94.75	0.95
90	3600	828.25	598.14	95.60	96.81	94.43	0.96
95	3800	5326.45	920.62	94.19	94.11	94.26	0.94

Figure 13 Training accuracy rates

4.5.4.2 Model building and training for the Legal Document Summarization Module

Model Architecture:

In the earlier sections, it has been pointed out that BART exhibits specific features conducive to the summarization process. Among these features is bi-directionality, which signifies the model's ability to comprehend text from both ends. Another significant feature is autoregression, a concept that deserves further clarification. Autoregression is a characteristic of the BART model wherein it generates text by progressively predicting and refining each token based on preceding tokens. Essentially, it can predict the next part of the text using what it has generated so far.

To tailor the pre-trained BART model for the task of summarizing legal case documents in the Sri Lankan context, a series of steps were taken:

- **Fine-tuning:** The researchers adjusted the BART model's parameters using the dataset comprising Sri Lankan legal cases. This process fine-tuned the model to better understand and summarize this specific type of legal data.
- **Tokenization:** As explained in [prior sections](#), tokenization was employed to prepare the legal text data for input into the model.
- **Hyperparameter tuning:** Modifications were made to certain model settings, including the learning rate, batch size, number of training epochs, and maximum sequence length. These adjustments were aimed at optimizing the model's performance for the unique task of summarizing legal case documents within the Sri Lankan legal domain.

The code snippet provided below offers a visual representation of these actions taken during the implementation process. Collectively, these steps ensure that the BART model is aptly configured to effectively summarize legal case documents in the Sri Lankan context, enhancing its efficiency and relevance.

```

def configure_optimizers(self):
    optimizer = AdamW(self.parameters(), lr=1e-5)
    return optimizer

model = LitModel(tokenizer, model=BartForConditionalGeneration.from_pretrained("facebook/bart-large"))

data_module = SummaryDataModule(tokenizer, dataset, batch_size=2)
trainer = pl.Trainer(max_epochs=50)
trainer.fit(model, data_module)

```

Figure 14 Hyperparameter tuning and fine tuning - Summarization model

Training procedure:

In the process of developing the text summarization model, a structured procedure is followed. It starts by setting up the model using the BART architecture, based on a pre-trained checkpoint. This pre-trained model serves as the foundation for the summarization task. The training data, which includes both the input text (referred to as 'context') and the target summary, are loaded. To prepare the data for training, tokenization is applied.

To make the training process efficient, the data is divided into batches, with the batch size determined by the "batch_size" hyperparameter. Batching allows for parallel processing on hardware like GPUs, which speeds up the training. Within each training epoch, the data is shuffled to prevent the model from learning patterns based on data order, which enhances its generalization.

For optimizing the model, the "AdamW" optimizer is utilized to adjust the model's weights during training. The learning rate (lr) hyperparameter controls the size of these weight updates. It's a crucial factor that affects training speed and stability. The model goes through multiple training epochs, as defined by "max_epochs," ensuring it learns from the entire dataset through several passes.

- **Hyperparameters:** There are two key hyperparameters in this model. They are: "batch_size" and "lr". The batch size determines how many data samples are processed at once during training. A larger batch size speeds up training but requires more memory. On the other hand, the learning rate controls the step size

in weight updates. Higher learning rates can lead to faster convergence but may be unstable, while lower rates offer stability but could be slower.

- **Data batching and shuffling:** Data batching involves dividing the training dataset into smaller groups for faster processing, and data shuffling randomizes the order of data samples within each training epoch to prevent the model from learning data order patterns.
- **Loss function:** The loss function is crucial in training the summarization model as it measures how different the model-generated summaries are from the target summaries. It's designed specifically for text generation tasks, quantifying the dissimilarity between predicted and actual summaries by comparing their token probabilities. During training, the goal is to minimize this loss by adjusting the model parameters using gradient-based optimization techniques. Essentially, the loss function guides the model toward generating high-quality summaries during training.

In summary, the development of the text summarization model involves setting up the model, preparing the data, optimizing with key hyperparameters, and training while closely monitoring the loss function to ensure the model produces accurate and meaningful summaries.

```
def training_step(self, batch, batch_idx):  
    input_encodings = batch[0]  
    label_encodings = batch[1]  
    outputs = self(input_encodings, labels=label_encodings)  
    loss = outputs.loss  
    return {'loss': loss}
```

Figure 15 Single training step - Summarization model

```
def train_dataloader(self):  
    return DataLoader(self.dataset, batch_size=self.batch_size, shuffle=True)
```

Figure 16 Enabling shuffling.

4.5.4.3 Model building and training for the Legal Question & Answering Module

At the core of the Legal Question Answering System, the essential components are the retriever model and the reader model. The retriever model efficiently scans vast legal document collections, while the reader model, a fine-tuned transformer-based architecture, excels in understanding complex legal language and extracting accurate answers. These components, meticulously trained and optimized, underpin the system's ability to deliver precise, context-aware responses to user queries.

4.5.4.3.1 Retriever model development

In the Legal Question Answering System, the retriever model plays a pivotal role in identifying and retrieving relevant documents. The choice of the Doc2Vec model, as opposed to traditional TF-IDF methods, offers distinct advantages.

Doc2Vec utilizes advanced neural network techniques, enabling it to capture semantic meaning and context in legal texts effectively. By generating document embeddings based on context and word order, Doc2Vec excels at understanding complex relationships and nuances within legal documents, ensuring precise retrieval for the system's optimal performance.

Retriever model training:

Selecting appropriate hyperparameters for training the Doc2Vec model plays a critical role in shaping the quality of document embeddings, and consequently, the overall performance of the retrieval system.

In the hyperparameter tuning process, the following values were carefully chosen: 50 epochs, a minimum word frequency (min_count) threshold of 10, a context window size (window) of 15, and an embedding dimensionality (vector_size) of 300.

- **Epochs:** A significant number of epochs (50) were selected to ensure the model captures intricate patterns within complex legal documents. Given the length and intricacy of legal texts, extended training helps the model grasp their structural nuances and semantics effectively.

- **Min_count:** This parameter, set at 10, strikes a balance by filtering out very rare terms while retaining relevant vocabulary. In legal documents, where rare legal terminology exists, this threshold ensures the model focuses on pertinent terms.
- **Window:** A larger window size (15) was chosen to allow the model to consider a broader context within sentences. In the realm of legal texts, where contextual understanding is paramount, this choice enhances the comprehension of complex legal language.
- **Vector_size:** A dimensionality of 300 for document embeddings was deemed suitable for capturing the intricate and multifaceted nature of legal information. This larger dimension enables the model to represent complex relationships and distinguish between contextually similar legal terms or concepts effectively.

Following the preprocessing and tagging of documents, the Doc2Vec model was trained using these defined hyperparameters, utilizing the Gensim Doc2Vec class. This meticulous approach ensures that the model's embeddings align optimally with the specific demands and complexities of legal document retrieval.

```
# Function to train Doc2Vec model
def train_doc2vec_model(documents):
    # Preprocess the document texts
    preprocessed_documents = [preprocess_text(document['documents']) for document in documents]

    # Tag the preprocessed documents
    tagged_documents = [TaggedDocument(doc, [i]) for i, doc in enumerate(preprocessed_documents)]

    # Train the Doc2Vec model
    model = Doc2Vec(tagged_documents, vector_size=300, window=15, min_count=10, epochs=50)

    return model
```

Figure 17 Training code segment - Doc2Vec model

Development of document retrieval function:

The retrieval of pertinent legal documents in response to queries was facilitated by the trained Doc2Vec model. The "doc2Vec_retrieve" function, illustrated below, utilizes the

input query, the trained model, and the desired number of document retrievals (k) to provide the most relevant documents based on similarity.

```
def doc2Vec_retrieve(question, documents, model, k):
    # Preprocess the question
    preprocessed_question = preprocess_text(question)

    # Infer the vector representation of the question using the trained Doc2Vec model
    question_vector = model.infer_vector(preprocessed_question)

    # Calculate the similarity scores between the question vector and document vectors
    similarity_scores = model.dv.most_similar([question_vector], topn=k)

    # Get the indices of the top-k documents based on the similarity scores
    top_indices = [i for i, _ in similarity_scores]

    # Retrieve the top-k documents (within the valid range)
    top_documents = [documents[i] for i in top_indices if i < len(documents)]

    return top_documents
```

Figure 18 Retriever code segment - Doc2Vec model

4.5.4.3.2 Reader model development

In the Legal Question Answering System, the reader model assumes a pivotal role in extracting answers from retrieved legal documents, serving as the primary tool for query interpretation and response. Given the outstanding performance of transformer-based models in NLP applications, RoBERTa, an optimized variant, was chosen for its exceptional accuracy and proficiency in comprehending intricate legal language nuances.

Reader model training:

The dataset, comprising 14,361 data points in SQuAD format (refer to the [Data preparation for the Legal Question & Answering Module](#)), underwent preprocessing via a Python function. This function tokenizes questions and contexts, determines answer span positions, and addresses out-of-context answers, crucial for training the model.

Then, for the training of the reader model, Figure 19 illustrates the selection of a pre-trained "RoBERTa-base" model from the "simpletransformers" library for fine-tuning and training on the prepared legal dataset. Through extensive research and testing,

hyperparameters were thoughtfully customized to strike a balance between performance and resource utilization, as depicted in Figure 20.

These carefully chosen hyperparameters include:

- **Max Sequence Length (max_seq_length):** Set at 128 to accommodate legal documents, controlling the maximum token count in each input sequence while considering potential memory limitations.
- **Number of Training Epochs (num_train_epochs):** Set at 50 to allow the model sufficient training time to grasp the intricacies of legal language and nuances.
- **Batch Sizes (train_batch_size and eval_batch_size):** Configured at 128 for training and 64 for evaluation to optimize memory utilization and training speed.
- **Evaluate During Training:** Enabled to monitor the model's performance during the training process, facilitating early issue detection when dealing with complex legal documents.
- **Evaluate During Training Steps:** Evaluation conducted every 1000 steps to provide frequent performance feedback, particularly valuable for challenging legal texts.
- **N Best Size (n_best_size):** Set to retain the top 3 best predictions, enhancing the model's ability to handle multiple possible answers within legal documents.
- **CUDA Usage (use_cuda):** Enabled to harness GPU acceleration, addressing the computational demands of a RoBERTa-based model.

The hyperparameter-tuned model underwent 50 epochs of training while continuously evaluating its performance. Throughout this training journey, the model acquired the capability to predict answer spans within legal documents based on associated questions, a vital component of its legal question-answering functionality.

```

import transformer libraries
import logging
from simpletransformers.question_answering import QuestionAnsweringModel, QuestionAnsweringArgs

#selection of the model type and model name
model_type="roberta"
model_name= "roberta-base"

```

Figure 19 Pretrained RoBERTa model import.

```

### Advanced Configure the Methodology
train_args = {
    "reprocess_input_data": True,
    "overwrite_output_dir": True,
    "use_cached_eval_features": True,
    "output_dir": f"outputs/{model_type}",
    "best_model_dir": f"outputs/{model_type}/best_model",
    "evaluate_during_training": True,
    "max_seq_length": 128,
    "num_train_epochs": 50,
    "evaluate_during_training_steps": 1000,
    "wandb_project": "Question Answer Application", #for training visualization
    "wandb_kwargs": {"name": model_name}, #for training visualization
    "save_model_every_epoch": False,
    "save_eval_checkpoints": False,
    "n_best_size": 3,
    # "use_early_stopping": True,
    # "early_stopping_metric": "mcc",
    # "n_gpu": 2,
    # "manual_seed": 4,
    # "use_multiprocessing": False,
    "train_batch_size": 128,
    "eval_batch_size": 64,
    # "config": {
    #     "output_hidden_states": True
    # }
}

```

Figure 20 Hyperparameter tuning RoBERTa model.

4.6 Implementation

The implementation of the Sri Lankan legal information retrieval system within this research project includes a multifaceted approach. It revolves around the creation of an extensive legal knowledge repository for end users utilizing a combination of Natural Language Processing (NLP), Named Entity Recognition (NER), and advanced Big Data technologies. These components collectively enable the effective organization, storage, and retrieval of vast legal information. The integration of AI models like BART and RoBERTa further enhances the system's capabilities, facilitating document summarization and question-answering to provide valuable legal insights. This holistic approach ensures

a comprehensive, user-friendly, and technologically advanced solution tailored to the intricacies of Sri Lanka's legal domain.

4.6.1 Data Extraction and Structuring Component

The Data Extraction and Structuring Component of the Sri Lanka Legal Information Retrieval System plays a pivotal role in the system's functionality and efficiency. This component is responsible for the systematic extraction of valuable legal information from a diverse range of sources. The implementation of this component involves employing advanced techniques for data extraction, such as natural language processing (NLP) where applicable, to transform unstructured legal texts into structured data formats.

4.6.1.1 Model integration

The Model Integration within the Data Structuring Module serves as a foundational element in the Sri Lanka Legal Information Retrieval System. When a user inserts a legal document, the built model takes center stage. Its primary role is to meticulously identify, and extract named entities embedded within the document. These identified entities are then seamlessly passed into a dedicated database, where they are cataloged for future reference and utilization. The database's contents are subsequently leveraged by both the Knowledge Graph Development Module and the Summarization Module, contributing vital structured data for these advanced components. In essence, the Data Structuring Module serves as the cornerstone of the system, acting as the base model upon which the entire system relies. It ensures that the legal information contained within documents is structured, organized, and readily available for knowledge graph development and summarization, thereby enhancing the overall efficiency and effectiveness of the Sri Lanka Legal Information Retrieval System.

4.6.2 Legal Knowledge Graph Development and Document Retrieval Component

The Legal Knowledge Graph (LKG) development and document retrieval component of this research project focuses on constructing a comprehensive knowledge graph using a multidimensional approach that leverages Natural Language Processing (NLP), Named Entity Recognition (NER), and Big Data technologies. The knowledge graph functions as

a fundamental component of the legal information retrieval system, facilitating the effective search and retrieval of case-report material obtained from judgments of the Sri Lankan Supreme Court. The methodology employed for this particular component is organized into a series of important steps. Those are as follows:

4.6.2.1 Defining relationships between identified legal entities

After accessing the stored legal entities from the Mongo DB database, this process starts the establishment of relationships among the identified legal entities. This is a crucial element in constructing the legal knowledge graph, which aims to accurately represent the whole of the knowledge base. The relationships shown in this component are meticulously constructed to represent the complex web of links and interdependencies existent within legal cases and entities as observed in Supreme Court rulings. Conceptual structures are utilized to facilitate the practical implementation of knowledge graphs.

In a theoretical structure, these linkages cover many aspects of a case. For example, the "INVOLVES" relationship connects a "CASE_NO" node to "PLAINTIFF" and "DEFENDANT" nodes, which represent the parties involved in a case. This theoretical foundation is consistent with the real-world legal context, which involves plaintiffs and defendants in every case. Similarly, the "REPRESENTED_BY" relationship resembles the legal representation of parties by counsel, simulating the scenario in which legal counsel advocates for plaintiffs and defendants.

In addition, the "DECIDED_BY" relationship connects a "CASE_NO" node to "JUDGE_1," "JUDGE_2," and "JUDGE_3" nodes, which represent the judges responsible for making decisions. This conceptualization reflects the crucial role that judges play in the legal process. In addition, the "BASED_ON" relationship connects "CASE_NO" nodes to "LEGAL_CONCEPT" nodes, emphasizing the legal principles upon which a case is based.

In real-world scenarios, the implementation of these connections commonly involves the use of a diverse range of advanced technologies. Tailored Cypher searches are utilized to establish knowledge graph relationships within the Neo4j graph database. The

employment of Python modules and scripts enhances the connection between legal case entities generated by the NER model and Cypher queries.

In addition, the practical implementation incorporates attributes within relationships, such as associating a particular case name with each relationship between entities. This attribute addition enhances the knowledge graph's practical utility by enabling the precise retrieval of case-specific information. Overall, the theoretical foundation of these relationships is consistent with legal domain knowledge, and their practical implementation leverages technology to create a knowledge graph that accurately represents the complexities of Supreme Court decisions. The combination of theory and practice ensures that the resulting knowledge graph is an effective tool for legal professionals and researchers, allowing for the efficient exploration and retrieval of legal knowledge within the Sri Lankan Supreme Court cases.

```
# Create relationships
relationships = [
    {"source_label": "CASE", "target_label": "PLAINTIFF", "type": "INVOLVES"},
    {"source_label": "CASE", "target_label": "DEFENDANT", "type": "INVOLVES"},
    {"source_label": "CASE", "target_label": "COUNSEL", "type": "REPRESENTED_BY"},
    {"source_label": "CASE", "target_label": "JUDGE1", "type": "DECIDED_BY"},
    {"source_label": "CASE", "target_label": "JUDGE2", "type": "DECIDED_BY"},
    {"source_label": "CASE", "target_label": "JUDGE3", "type": "DECIDED_BY"},
    {"source_label": "CASE", "target_label": "LEGAL_CONCEPT", "type": "BASED_ON"},
    {"source_label": "CASE", "target_label": "LEGAL_CASE_REFERENCE", "type": "HAS_REFERENCE"},
    {"source_label": "CASE", "target_label": "ARGUED_ON", "type": "ARGUED_ON"},
    {"source_label": "CASE", "target_label": "DECIDED_ON", "type": "DECIDED_ON"},
    {"source_label": "JUDGE1", "target_label": "JUDGE1_DECISION", "type": "MADE_DECISION"},
    {"source_label": "JUDGE2", "target_label": "JUDGE2_DECISION", "type": "MADE_DECISION"},
    {"source_label": "JUDGE3", "target_label": "JUDGE3_DECISION", "type": "MADE_DECISION"},
    {"source_label": "PLAINTIFF", "target_label": "COUNSEL", "type": "REPRESENTED_BY"},
    {"source_label": "DEFENDANT", "target_label": "COUNSEL", "type": "REPRESENTED_BY"},
    {"source_label": "CASE", "target_label": "OTHER_PARTIES", "type": "INVOLVES"},
]
```

Figure 21 Defining the relationships found in each case between legal entities

4.6.2.2 Defining nodes & edges and constructing the knowledge graph

Once legal entities have been accurately identified and their relationships defined, the subsequent important stage involves the definition of nodes and edges. In this phase, the nodes are precisely defined to represent the legal entities, while the edges are specified to represent the relationships between them. Every identified relationship is further characterized by attributes, such as entity names, which increases the knowledge graph's

ability to reflect the intricate network of legal connections and interconnections. This stage guarantees that the resulting graph precisely represents the interrelationships of the legal cases.

The utilization of Neo4j, a prominent graph database technology in the industry, is integral to the construction of the knowledge graph, which serves as the core component of the technique implemented in this component. Neo4j is a highly efficient platform that is well-suited for the management of extracted legal entities and their relationships within a well-defined and organized structure. The knowledge graph is systematically constructed via Cypher queries, which is a query language designed by Neo4j specifically for graph databases. The graph shown above serves as a complete compilation of legal knowledge, characterized by a linked structure that facilitates effortless exploration and exhaustive analysis. The effectiveness of Neo4j's relationship management allows the precise representation of legal links inside the knowledge graph, encompassing their intricate complexity.

The significance of this phase lies in its ability to convert the recognized legal case entities (case information) and their interconnections into a resource that is easily accessible and navigatable, catering to the needs of legal practitioners, attorneys, and law students within the Sri Lankan legal sector. The knowledge graph that was built serves as a robust tool for accessing legal case information, obtaining legal viewpoints, and comprehending the complex legal landscape of Supreme Court rulings in Sri Lanka.

```
# Create nodes
for entity in entities:
    label = entity['label']
    node_name = entity['text']
    formatted_label = label.replace(" ", "_") # Replace the space with an underscore
    session.run(f"MERGE (n:{formatted_label} {{name: $name}})", name=node_name)
```

Figure 22 Code Snapshot for node creation for each legal entity found in cases

```

# Create relationships
for rel in relationships:
    source_formatted_label = rel['source_label'].replace(" ", "_")
    target_formatted_label = rel['target_label'].replace(" ", "_")

    # Check if both source and target entities exist in the 'entities' list
    source_entity = next((e for e in entities if e['label'] == rel["source_label"]), None)
    target_entity = next((e for e in entities if e['label'] == rel["target_label"]), None)

    if source_entity and target_entity:
        source_name = source_entity['text']
        target_name = target_entity['text']

        query = f"""
        MATCH (a:{source_formatted_label}), (b:{target_formatted_label})
        WHERE a.name = $source_name AND b.name = $target_name
        MERGE (a)-[r:{rel['type']}]->(b)
        """

        if rel['type'] == "MADE_DECISION":
            case_entity = next((e for e in entities if e['label'] == "CASE"), None)
            case_name = case_entity['text'] if case_entity else None
            query += " SET r.case_name = $case_name"
        else:
            case_entity = next((e for e in entities if e['label'] == "CASE"), None)
            case_name = case_entity['text'] if case_entity else None
            query += " SET r.case_name = $case_name"

        session.run(query, source_name=source_name, target_name=target_name, case_name=case_name)
    else:
        print(f"One or more entities are missing for the relationship: {rel}. Skipping relationship creation.")

```

Figure 23 Code Snapshot for edges creation for each relationship between entities

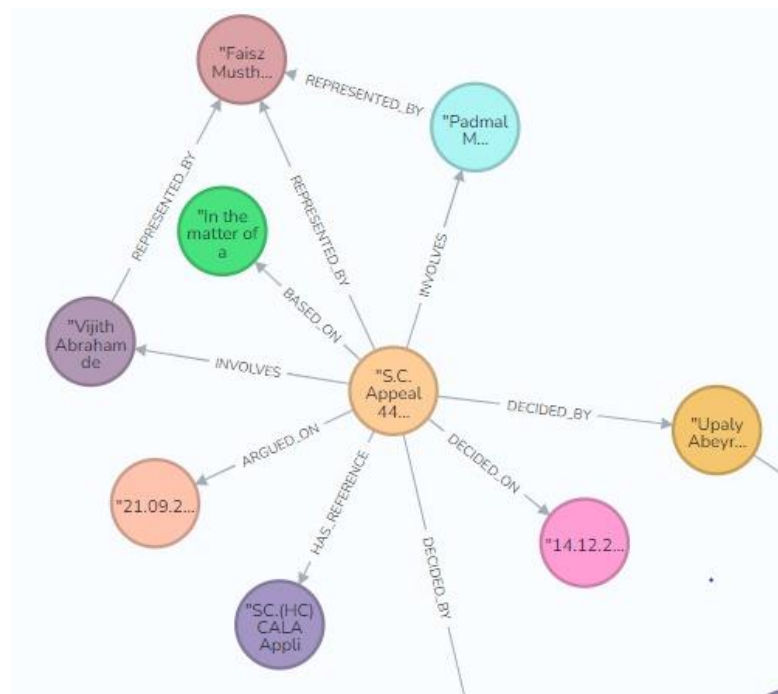


Figure 24 Part of the built knowledge graph with nodes and edges on Neo4j

4.6.2.3 Cypher query implementation

Then the component employs the Cypher query language and a proficient graph database tool, such as Neo4j, to facilitate user interaction with the knowledge graph. Cypher queries facilitate the analysis and retrieval of legal information from the knowledge graph. The knowledge graph has a high degree of user-friendliness and adaptability, effectively catering to the requirements of legal professionals, attorneys, and individuals pursuing legal education. Cypher searches can efficiently and precisely retrieve information, irrespective of whether the material relates to a specific judge, a particular legal concept, or cases argued on specified dates or any other search criteria.

The utilization of Cypher queries also facilitates the extraction of legal insights from the knowledge graph. Supreme Court opinions contain hidden patterns, trends, and relationships that can be discovered by end users. The utilization of analytical skills aids lawyers, law students, and practitioners in enhancing their arguments, cases, and understanding of legal precedents. Additionally, Cypher queries enhance the user experience by examining the intricate legal connections inside the knowledge graph. This allows for a comprehensive understanding of the legal history of a case, the various parties involved, and the growth of legal concepts within cases.

The utilization of Cypher queries enables the realization of the complete potential of the knowledge network, transforming it from a static repository of data into a dynamic and interactive resource. The simplification of Sri Lankan Supreme Court judgments renders the legal information retrieval system essential for legal professionals and scholars.

```
1 MATCH (n)-[r]→(m)
2 RETURN n, r, m
```

Figure 25 Cypher query for retrieving all cases from a knowledge graph

```
1 MATCH (n)
2 DETACH DELETE n
```

Figure 26 Cypher query to delete whole knowledge graph (all the cases)

```

1 MATCH (c:CASE_NO {name: "SC/2013"})-[r]-()
2 DETACH DELETE c, r

```

Figure 27 Cypher query for deleting a specific case

4.6.2.4 User interface development and integration with the knowledge graph

Along with the development of the knowledge graph, a user interface is systematically designed and developed. The web application interface offers customers a user-friendly way to access the case information from the knowledge graph. The system offers advanced filtering options, including case numbers, involved parties, dates, judges, legal citations, and other essential characteristics. The interface provided to users is designed to be user-friendly, allowing for precise searches of legal cases. This feature enhances the user experience by aiding the proper retrieval of pertinent legal material. The user interface functions as the primary interface of the legal information retrieval system, facilitating the connection between users and the extensive collection of legal case knowledge stored within the graph.

Case	Legal Case References	Legal Concepts	Plaintiff	Defendant	Other Parties
SC Appeal No: 15/2013	CP/HCCA/KAN/61/2011 Vs.	In the matter of a Special Leave to Appeal in terms of Article 154(P) read with the Constitution read	Jathika Sevaka Sangamaya, No. 416, Kotte Road, Pilakotte, (On behalf of S.S. Samarasinghe, G.V.A.N. Senadheera, S.N. Nanay)	Sri Lanka Hadabima Authority No. 08, Gonnoruwa Road, Peradeniya.	Jathika Sevaka Sangamaya

Figure 28 Front end integration with Knowledge Graph

4.6.3 Legal Case Summarization Component

Following the successful implementation of the specialized legal domain summarization model tailored for Sri Lanka, the next important step is to smoothly add the BART model, which was created, to the system's user interface. This means making sure the model works well, within the system so that users can use it easily.

4.6.3.1 Model saving

To achieve this integration, the model must be preserved. Once the model has undergone the training process to generate concise legal document summaries, it is important to save the model for future utilization within the system's front end. This strategic step holds the potential to significantly reduce response times for user queries, thus enhancing overall user experience. The accompanying figure below offers a visual representation of the code segment employed for model preservation.

```
import shutil

# Define the paths
model_dir = "/content/sample_data/model_summarization" # Path to the saved model directory
zip_filename = "/content/sample_data/model_summarization.zip" # Path for the ZIP file

# Save the model and tokenizer
model.model.save_pretrained(model_dir)
tokenizer.save_pretrained(model_dir)
```

Figure 29 Code Snapshot for Model Saving - BART

By incorporating this functionality, the legal information retrieval system will operate with increased efficiency, enabling users to access summarizations swiftly and effectively, ultimately contributing to the system's overall utility and accessibility in the legal domain of Sri Lanka.

4.6.3.2 Frontend implementation

During the development of the system's front end, a significant consideration revolved around the user base. The team recognized that the system caters to individuals with demanding schedules within the legal field, and they prioritized simplicity in the user interface design. This approach aimed to ensure accessibility for busy legal professionals.

In the provided figure, the User Interface (UI) of the summarization component is presented. The front-end implementation of this application utilizes "Flask" as its framework.

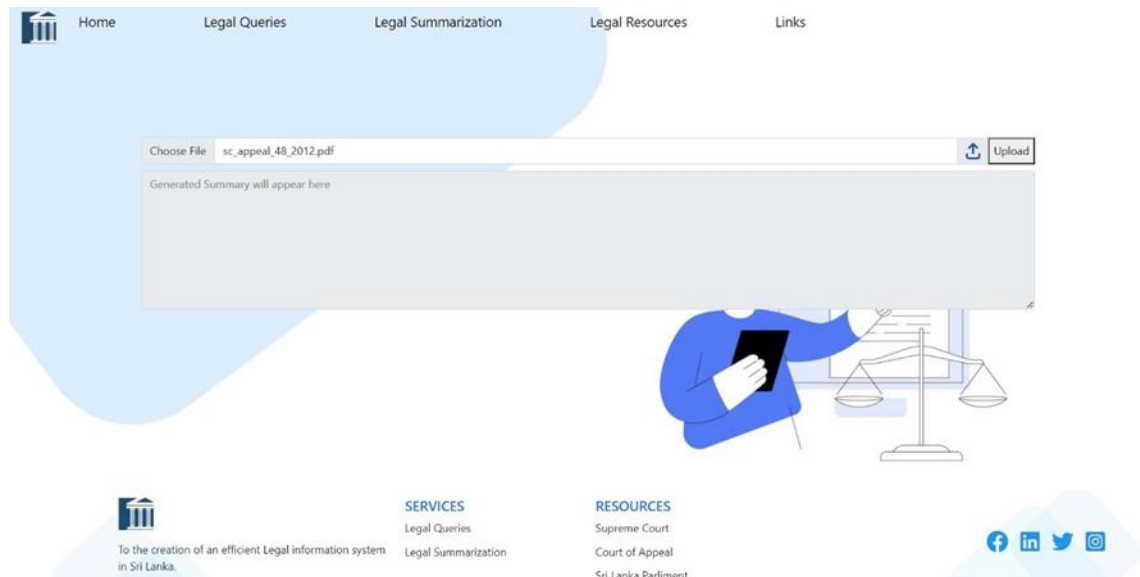


Figure 30 Summarization module UI design

The user interface allows a straightforward interaction process. Users can select a document by clicking on the "Choose File" button and proceed to upload it using the "Upload" button. At this point, the chosen document undergoes parsing through the Custom Named Entity Recognition (NER) model, which is hosted in the backend. The NER model extracts the relevant judgment section from the document.

Subsequently, the extracted judgment section is passed through the summarization model, also hosted in the backend. This model processes the content and generates a concise summary. The final output is then presented to the user through the interface.

For further insight into the system's functionality, interested parties may refer to the accompanying figures below, which provide visual snapshots illustrating each of the aforementioned steps.

```

# Load BART model and tokenizer
model_dir = "model_summarization"
bart_tokenizer = BartTokenizer.from_pretrained(model_dir)
bart_model = BartForConditionalGeneration.from_pretrained(model_dir)

# Load spaCy NER model
nlp_ner = spacy.load("model-best")

```

Figure 31 Loading the summarization model to the frontend.

```

@app.route('/summarization', methods=['GET', 'POST'])
def summarization():
    if request.method == 'POST':
        file = request.files['pdf_file']
        # Read the contents of the uploaded PDF file
        text = ""
        try:
            with pdfplumber.open(file) as pdf:
                for page in pdf.pages:
                    text += page.extract_text()
        except:
            return "Error parsing the PDF file"

        # Extract entities using spaCy NER model
        doc = nlp_ner(text)

        # Find the entity labeled as "JUDGMENT_1"
        extracted_entity = None
        for ent in doc.ents:
            if ent.label_ == "JUDGMENT_1":
                extracted_entity = ent.text
                break

```

Figure 32 Routing of the summarization model frontend

4.6.4 Legal Question & Answering Component

The implementation phase of the Question-Answer System comprises two pivotal stages. Initially, it involves the creation of the question-answering pipeline, which is subsequently

followed by the integration process facilitating seamless communication between the frontend and backend components.

4.6.4.1 Building the question-answering pipeline

The initial phase of implementation centers on establishing a meticulously designed question-answering pipeline, carefully crafted to streamline the processing of user queries and the extraction of precise answers from legal documents.

Loading of saved models:

Following the successful training and optimization of both the retriever and reader models, a strategic step involved saving these models in a designated storage location. This approach ensures expedient access to the models whenever a user query is submitted, facilitating efficient loading and utilization within the system's operational framework.

Model integration:

Having established both the retriever and reader models, the subsequent step involves the construction of the question-answering pipeline, as depicted in the diagram below. This integrated system is carefully designed to facilitate the smooth processing of user queries, following a coherent sequence of actions. Initially, the retriever model comes into play, discerning and identifying the most pertinent document within the corpus. Subsequently, this selected document serves as the contextual basis for the reader model, which is responsible for extracting precise answers to the user's questions. This well-structured pipeline ensures the efficient functioning of the system, enabling it to deliver accurate responses to user queries.

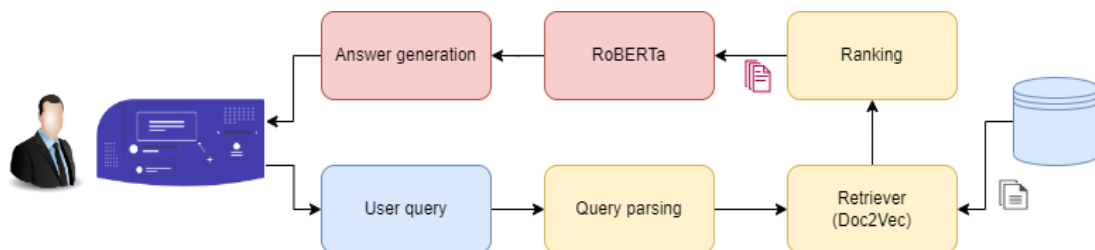


Figure 33 QnA pipeline

4.6.4.2 Implementation of frontend and integration

The seamless integration of the Legal Question Answering System with its frontend interface holds paramount importance, particularly due to the wide-ranging user base characterized by varying levels of technical proficiencies. The user interface boasts an intuitively crafted design, as visually depicted in the accompanying illustrative representation. It delivers a user-centric interaction experience, enabling users of diverse backgrounds to effortlessly input their queries and swiftly access legal information straightforwardly and efficiently.

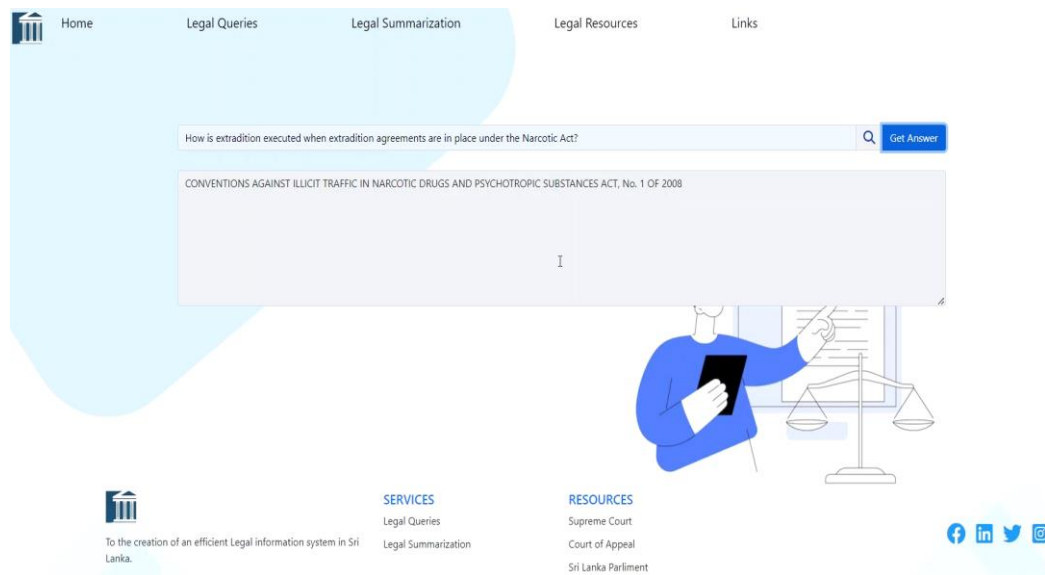


Figure 34 Question answering system UI

The system has been meticulously constructed as a web application, with Python Flask serving as the basic development framework. A key aspect of the user interface is the text box, shown in the illustration above, which allows users to enter their legal queries.

Subsequently, this text field acts as the gateway, channeling the user's query into the question-answering pipeline. Within this well-orchestrated pipeline, the query undergoes the intricate process outlined earlier, to extract precise and tailored answers. Upon successfully extracting the answer, the system seamlessly employs Flask routing mechanisms to relay the answer back to the user interface, as depicted in the figure below.

```
if answer_document:
    # Pass both the answer and the document name to the frontend
    return render_template("legalQuaries.html", question=query_question, answer=answer, document_name=answer_document['title'])
else:
    return render_template("legalQuaries.html", question=query_question, answer="No answer document found.")
```

Figure 35 Answer routing to UI

This elegant design and routing architecture ensure a user-friendly experience, allowing users to effortlessly interact with the system while obtaining accurate and timely legal information.

4.7 Testing

The System testing phase is a crucial step in ensuring the robustness and dependability of the system. The testing methodology includes multiple levels of testing, each serving a different purpose and contributing to the Sri Lankan information retrieval system's quality. This section presents an overview of the testing approach. The testing approach specifies the elements to be tested, considering function relevance and vulnerability to end users.

4.7.1 Test planning

The implementation of structured testing is vital to monitor the progress of a project and determine which types of tests are necessary. The testing methodology defines the components that are to be subjected to testing, taking into account their functional significance and vulnerability to exploitation by end users. Test cases were created based on available use cases and executed manually, with outcomes methodically documented.

Test strategy:

1. Define the component to be tested.
2. Select functions based on priority and user vulnerability.
3. Execute the test plan.
4. Record results manually and automatically.
5. Identify and correct errors.

6. Repeat test cases until the expected outcome is achieved.

4.7.2 Test case design

Test cases were developed to validate all the components' functionality. The following test cases were executed to verify the functionality and accuracy of each component:

4.7.2.1 Unit testing

Individual components of the Sri Lankan Legal information retrieval system are tested to verify their correctness.

Data Extraction & Structuring component:

Table 3 Unit testing 01.

Test Case ID	AYCA-01
Test Title	Legal Entity Identification
Test Case Objective	To verify that the NLP and Named Entity Recognition (NER) processes correctly identify legal entities from Supreme Court verdict documents.
Test Steps	1. Input a sample Supreme Court verdict document. 2. Execute the entity extraction process. 3. Check if legal entities, such as case numbers, judges' names, legal citations, argued date, decided date, judgments, counsel, plaintiff, defendant, and other parties are accurately identified.
Expected Result	The entity extraction process correctly identifies all legal entities.

Actual Result	
Pass/Fail	Pass

Legal Knowledge Graph Development and Document Retrieval Component:

Table 4 Unit testing 02

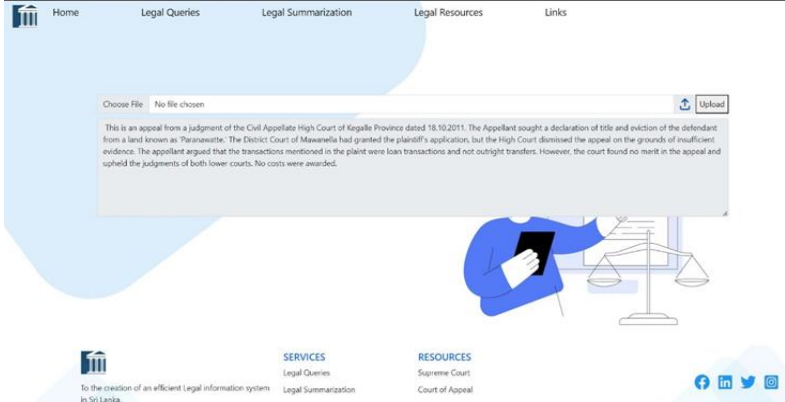
Test Case ID	AYCA-02
Test Title	Relationship Identification & Definition
Test Case Objective	To ensure that the relationships between legal entities are correctly identified and represented.
Test Steps	1. Input a sample set of legal entities. 2. Execute the relationship identification & definition process. 3. Verify that relationships, such as “(CASE_NO)- [: INVOLVES] ->(PLAINTIFF) (CASE_NO)- [: INVOLVES] ->(DEFENDANT) (CASE_NO)- [: INVOLVES] ->(OTHER_PARTIES) (CASE_NO)- [: ARGUED_ON] ->(ARGUED_ON_DATE) (CASE_NO)- [: REPRESENTED_BY] ->(COUNSEL) (CASE_NO)- [: DECIDED_ON] ->(DECIDED_ON_DATE) (CASE_NO)- [: DECIDED_BY] ->(JUDGE_1) (CASE_NO)- [: DECIDED_BY] ->(JUDGE_2)

	(CASE_NO)- [: DECIDED_BY] ->(JUDGE_3) (JUDGE_1)- [: MADE_DECISION] ->(JUDGEMENT_1) (JUDGE_2)- [: MADE_DECISION]->(JUDGEMENT_2) (JUDGE_3)- [: MADE_DECISION]->(JUDGEMENT_3) (CASE_NO)- [: BASED_ON] ->(LEGAL_CONCEPT) (CASE_NO)- [: HAS_REFERENCE] - >(LEGAL_CASE_REFERENCE) “ are correctly established.
Expected Result	The relationship identification & definition process correctly establishes relationships between legal entities.
Actual Result	
Pass/Fail	Pass

Legal Document Summarization component:

Table 5 Unit testing 03

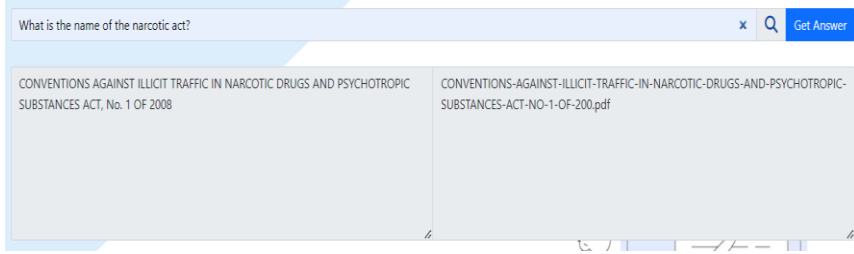
Test Case ID	AYCA-03
Test Title	Legal document summarizer gives a comprehensive summary.

Test Case Objective	To ensure that a comprehensive summary is given.
Test Steps	1. Input a sample document of a case file. 2. Execute the summarization model. 3. Get a comprehensive summary.
Expected Result	A concise summary of the given judgment for the case document sc_appeal_48_2012.
Actual Result	
Pass/Fail	Pass

Legal Question Answering component:

Table 6 Unit testing 04

Test Case ID	AYCA-04
Test Title	Reader model test for answerable question
Test Case Objective	To ensure that the answer is correct for the question.
Test Steps	1. Input a sample set of questions based on the Narcotic Drugs Act. What is the name of the narcotic act?

Expected Result	CONVENTIONS AGAINST ILLICIT TRAFFIC IN NARCOTIC DRUGS AND PSYCHOTROPIC SUBSTANCES ACT, No. 1 OF 2008
Actual Result	
Pass/Fail	Pass

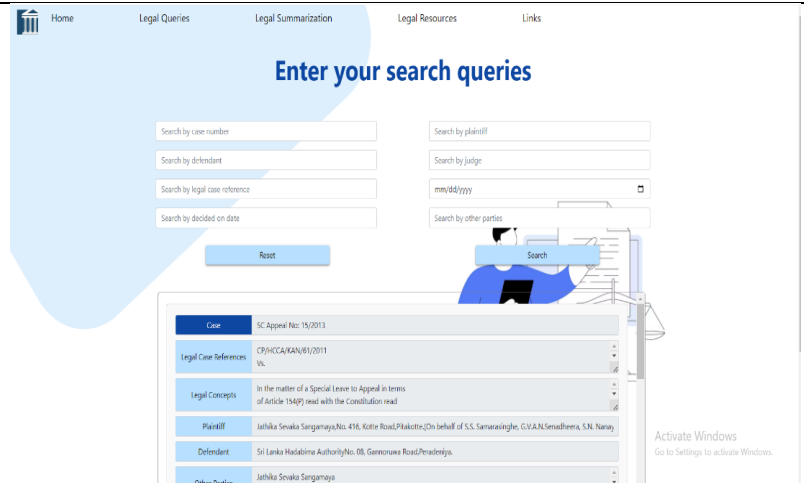
4.7.2.2 Integration Testing

Testing was conducted to ensure proper communication between the front-end and back-end components. This includes verifying that user queries are correctly routed to the pipeline and that answers are displayed accurately in the user interface.

Legal Knowledge Graph Development and Document Retrieval component:

Table 7 Integration testing 01

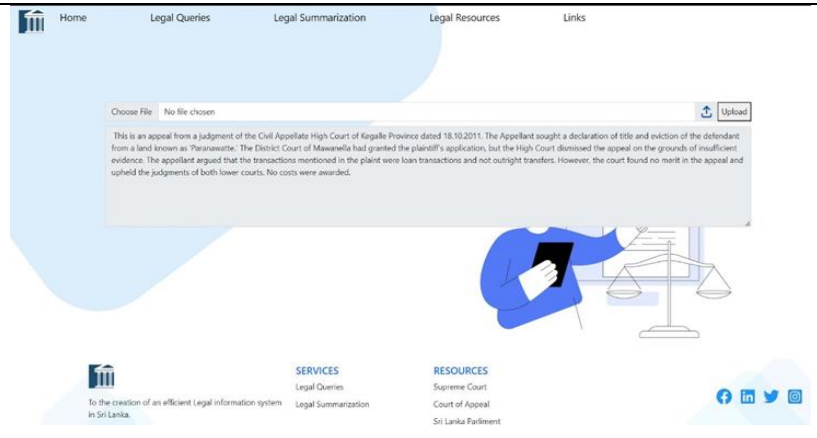
Test Case ID	AYCA-05
Test Title	Retrieve case information through user interface interaction
Test Case Objective	To verify that the user interface allows users to interact effectively with the knowledge graph.
Test Steps	1. Access the user interface. 2. Input specific query criteria, such as a case number or judge's name. 3. Execute queries through the interface. 4. Verify if the interface provides accurate results and options for further exploration.

Expected Result	The user interface enables users to interact seamlessly with the knowledge graph and retrieve relevant legal case information as per search criteria.
Actual Result	
Pass/Fail	Pass

Legal Document Summarization component:

Table 8 Integration testing 02

Test Case ID	AYCA-06
Test Title	Retrieve case summary through user interface interaction.
Test Case Objective	When the user inputs a legal case document, he/she needs to get the summary.
Test Steps	1. Access the user interface. 2. Insert the legal case document “sc_appeal_48_2012.”
Expected Result	A concise insightful summary of the given judgment for the case document sc_appeal_48_2012.

Actual Result	
Pass/Fail	Pass

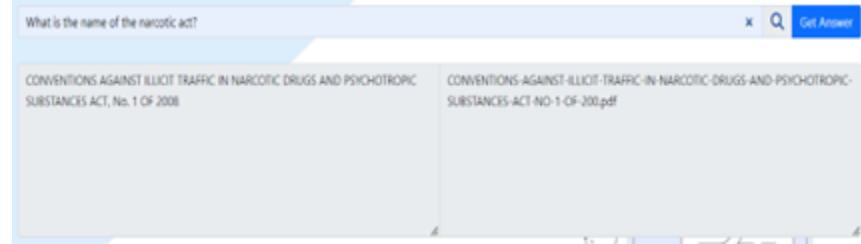
Legal Question Answering component:

Table 9 Integration testing 03

Test Case ID	AYCA-07
Test Title	Retrieve long-form legal answers through user interface interaction.
Test Case Objective	Testing long-form answer extraction for a question.
Test Steps	1. Access the user interface. 2. Type the question “What are the SAARC countries?”
Expected Result	Answer: 1. Peoples Republic of Bangladesh 2. Kingdom of Bhutan 3. Republic of India 4. Republic of Maldives 5. Kingdom of Nepal 6. Islamic Republic of Pakistan 7. Islamic Republic of Afghanistan Document: Narcotic Act No. 1 of 2008 pdf Document

Actual Result	
Pass/Fail	Pass

Table 10 Integration testing 04

Test Case ID	AYCA-08
Test Title	Retrieve short-form legal answers through user interface interaction.
Test Case Objective	Testing short-form answer extraction for a question.
Test Steps	1. Access the user interface. 2. Type the question “What is the name of the narcotic act?”
Expected Result	Answer: CONVENTIONS AGAINST ILLICIT TRAFFIC IN NARCOTIC DRUGS AND PSYCHOTROPIC SUBSTANCES ACT, No. 1 OF 2008 Document: Narcotic Act No. 1 of 2008 pdf Document
Actual Result	
Pass/Fail	Pass

4.7.2.3 User acceptance testing (UAT)

The process of usability testing involves the active participation of genuine users who act as representatives of the target client community. The evaluation covered the user-friendliness of the system, the design of its interface, and the overall user experience. Subsequently, comments were provided.

The User Acceptance Testing (UAT) phase comprised the active involvement of real users, including students from Sri Lanka Law College, and an external industry specialist who was also the External Supervisor/client.

The confirmation of the system's compliance with requirements and effective delivery was confirmed by an official letter, which can be found in [APPENDIX B](#).

4.8 Commercialization of the Product

The legal industry in Sri Lanka stands as one of the most profitable and financially stable sectors of the country. Its vast landscape offers abundant financial opportunities, waiting to be harnessed by innovative products that aim to streamline and enhance this domain.

To enhance the product's appeal to stakeholders and pave the way for its commercialization, a marketable name, 'AYCA', has been assigned.



Figure 36 Product logo

The 'AYCA' system, bearing the slogan 'We Democratize Law,' resonates with the core values and objectives of the system and represents the initial step toward commercializing this innovative solution within the legal industry. It is poised to unlock new dimensions in legal information retrieval and accessibility, ultimately benefiting legal professionals, students, and the broader public.

4.8.1 Market opportunity

Market research was conducted to gauge the marketability of the 'AYCA' system. A survey, involving over 170 participants, revealed key insights into the potential user base and demand for the product. The survey results underscore that the market for the system is composed of three primary audiences: legal professionals, law students, and the general public. Among these, legal professionals emerge as the primary audience, with law students and the public following closely.

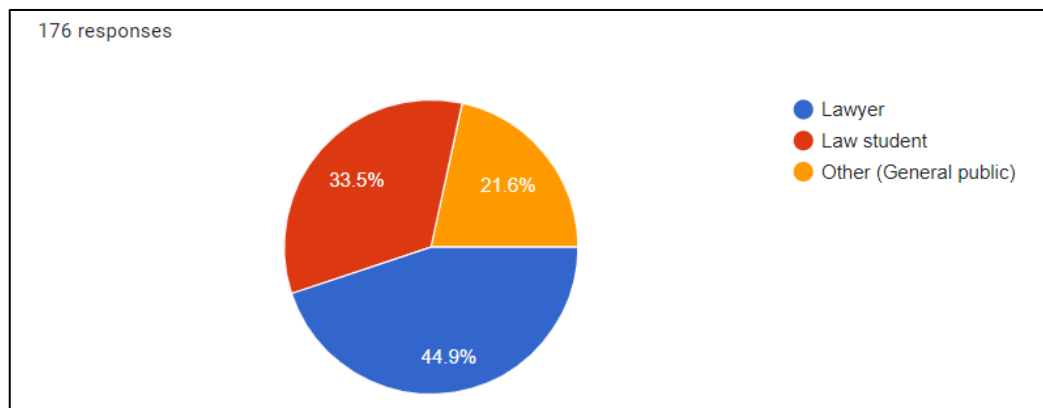


Figure 37 Target audience

Remarkably, the survey highlighted a compelling and immediate need for a comprehensive legal information retrieval system within the country. An overwhelming majority of the participants (95.4%) concurred that there exists a substantial demand for such a system. Furthermore, a significant portion expressed their willingness to transition to or utilize such a system if it were to be developed (refer to [APPENDIX A](#) for survey results).

The findings from the market research unequivocally indicate the presence of a robust market for the proposed 'AYCA' system. Moreover, the survey demonstrates that the target audiences are not only receptive but also eager to embrace such a product should it be introduced to the market. These insights underscore the system's potential for commercial success and its capacity to meet the pressing needs of the legal community and the broader public.

4.8.2 Market competition

Market competition plays a pivotal role in determining a product's viability. In the case of the Sri Lankan Legal Information Retrieval System, 'AYCA,' market penetrability is notably high due to the absence of formidable competitors within the Sri Lankan legal domain.

Currently, the market is primarily served by two systems. The first is a government-maintained catalog that offers only rudimentary legal document organization and a basic search function, falling far short of meeting the comprehensive needs of legal professionals. The second system, a premium offering system, shares similar limitations and lacks the advanced functionalities offered by 'AYCA.' Consequently, the proposed system enjoys a significant technological and functional advantage over existing competition.

In summary, the survey results (refer to [APPENDIX A](#) for survey results), combined with the evident technological gap in the market, affirm that stakeholders are not only receptive but also eager to transition to the 'AYCA' system once it is commercially available. The absence of comparable alternatives positions 'AYCA' as a pioneering solution poised to address the pressing demands of the legal community and offer unmatched functionality in the Sri Lankan legal information retrieval landscape.

4.8.3 Commercialization strategy

Target group coverage:

The market strategy for 'AYCA' involves catering to diverse user groups while adhering to the core principle of democratizing legal knowledge.

- To effectively target legal professionals, a premium version of the product will be introduced. This premium offering will be available as a standalone solution for solo practitioners and as a package tailored for legal firms. This strategic approach ensures that the primary audience, legal professionals, can access advanced features and functionalities tailored to their specific needs.
- Recognizing the financial constraints of students, 'AYCA' will also offer specialized packages to legal education institutions. These packages will provide cost-effective access to the system, enabling students to benefit from its extensive capabilities. This approach ensures that the legal education sector, a crucial user base, can integrate 'AYCA' into their curriculum and research activities without financial constraints.
- Moreover, in line with the system's core value of democratizing legal knowledge, a free version with limited functionality will be made available to the general public. This offering ensures that legal information remains accessible to all, aligning with the broader goal of promoting legal literacy and understanding within Sri Lanka.

Through this multifaceted market strategy, 'AYCA' seeks to cater to the diverse needs of its user base while fostering inclusivity in legal information access.

Commercialization plan:

The commercialization plan for the system encompasses a strategic three-year timeline, meticulously designed to address distinct phases of market development.

- **Year 1 - Market Expansion (Aggressive Market Capture):** In the inaugural year, the system will be introduced into the market with a keen focus on aggressive

market capture. The pricing strategy during this phase will be highly competitive, aimed at swiftly gaining a substantial market share. The primary objective is to expand the user base rapidly, prioritizing market penetration over immediate profitability. Breaking even will be the fundamental financial goal for this phase.

- **Year 2 - Market Stabilization:** Building upon the foundation established in the first year, the second year will emphasize market stabilization. The system will be offered at a flat rate to consumers, ensuring consistency and reliability of service. While profitability remains a consideration, the primary focus will be on maintaining a strong market position and ensuring customer satisfaction. This phase aims to solidify the system's presence and reputation within the market.
- **Year 3 Onwards - Profitability Emphasis:** Starting from the third year and extending into subsequent years, the primary emphasis will shift towards profitability. A thorough reevaluation of the pricing model will occur, introducing diversified offerings such as packaged models and pay-as-you-go services. This phase aims to optimize revenue generation while continuing to provide exceptional value to customers. Profitability becomes a central goal while upholding the commitment to meeting the evolving needs of users.

This meticulously phased approach reflects a well-structured commercialization plan, intelligently adapting to the different stages of product maturity and market development. It ensures that 'AYCA' not only establishes a robust presence but also evolves into a sustainable and profitable venture over time.

5 RESULTS & DISCUSSION

The primary outcomes and discoveries of the research are presented and examined in the subsequent sections. The discussion part provides a comprehensive analysis and elaboration of the results and conclusions.

5.1 Results

5.1.1 Data Extraction and Structuring Component

The evaluation of the legal structuring module's performance is conducted using fundamental evaluation metrics like recall, precision, and F1-score. These metrics serve as essential benchmarks to gauge the system's effectiveness in accurately extracting and structuring legal information.

- Recall measures the module's ability to capture all relevant legal entities, ensuring comprehensive coverage.
- Precision assesses the precision of the module by examining the correctness of the extracted entities, minimizing false positives.
- The F1-score provides a balanced evaluation, considering both recall and precision, offering a holistic view of the system's overall accuracy.

Table 11 Data extraction model performance

Model	Recall	Precision	F1 Score	Accuracy score
spaCy	94.12	94.11	94.19	0.94

5.1.2 Knowledge Graph Development and Document Retrieval Component

The incorporation of the Knowledge Graph Development and Document Retrieval components in this research project has brought about an important improvement in the field of legal information retrieval, particularly in the context of retrieving Supreme Court Verdict information in the Sri Lankan legal domain. By employing a thorough and precisely planned methodology, this approach has successfully utilized Natural Language

Processing (NLP), Named Entity Recognition (NER), and Neo4j graph database technology to create a robust knowledge graph that accurately represents the system's knowledge base. The creation of this knowledge graph serves as evidence of the approach's ability to offer a thorough and interconnected representation of legal entities and their complex relationships within the collection of Sri Lankan Supreme Court judgments and cases.

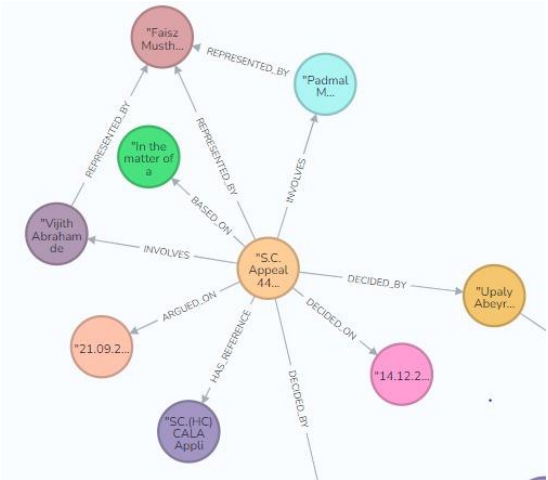


Figure 38 Part of built knowledge graph on neo4j

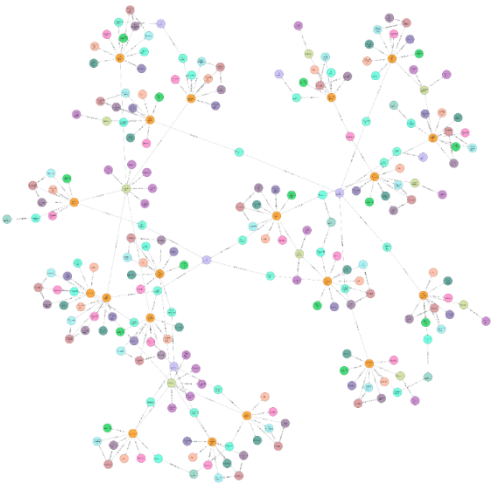


Figure 39 Constructed full knowledge graph with all cases on neo4j graph storage

Performance measures of the constructed knowledge graph:

To assess the efficiency and effectiveness of information retrieval from the created knowledge graph, a thorough series of performance evaluations was systematically conducted, encompassing a range of critical performance characteristics. These performance indicators provide a thorough comprehension of the capabilities and optimizations achieved within the system. The subsequent factors are the principal performance indicators and their accompanying implications:

To evaluate the performance of the Knowledge graph, a series of Cypher queries (Cypher being a querying language designed for Neo4j graph storage) were conducted. These queries aimed to obtain case information from the constructed knowledge graph based on the search parameters specified by end users.

- **Average Query Execution Time:** The approach given in this study showcases outstanding query execution performance, as seen by a remarkably low average query execution time of approximately 179 milliseconds. The exceptional efficacy of this approach ensures that legal experts can efficiently access the required information, significantly enhancing research effectiveness.

$$\text{Average Query Execution Time} = \frac{\text{Total Query Execution Time (in milliseconds)}}{\text{Number of Queries Executed}}$$

- **Memory Usage (Per Query):** The aforementioned methodology demonstrates efficient resource consumption by efficiently maintaining a conservative memory usage of approximately 25,638 bytes per query. This ensures the efficient operation of the knowledge graph while minimizing the strain on hardware resources.

$$\text{Memory Usage (Per Query)} = \frac{\text{Total Memory Usage (in bytes)}}{\text{Number of Queries Executed}}$$

- **Page Cache Hit (Per Query):** A substantial average of around 1000.5 page cache hits per query results in a significant decrease in disk reads. Hence, enhancing the

efficiency of data retrieval. The performance metric demonstrates the efficacy of the knowledge graph in retrieving data from its cache, mitigating delay.

Page Cache Hit (Per Query) = Total Page cache Hits / Number of Queries Executed

- **Page Cache Misses (Per Query):** The extremely low occurrence of page cache misses provides proof of the exceptional efficiency in utilizing the cache within our knowledge network. The utilization of the optimization technique results in reduced delays caused by disk reads, hence significantly enhancing the efficiency of data retrieval.

Page Cache Misses (Per Query) = Total Page cache Misses / Number of Queries Executed

- **Database Hits (Per Query):** The knowledge graph demonstrates its robust retrieval capabilities by achieving an average of around 3533.5 database hits per query. This metric demonstrates the ability of the knowledge graph to effectively and precisely retrieve an extensive range of legal data.

Database Hits (Per Query) = Total Database Hits / Number of Queries Executed

Table 12 Knowledge graph performance

Metric	Average Value
Average Query Execution Time	179 ms
Memory Usage (per query)	25,638 Bytes
Page Cache Hits (per query)	1000.5
Page Cache Misses (per query)	0
Estimated Rows (per query)	Varies (39-113)
Database Hits (per query)	3533.5

5.1.3 Legal Document Summarization Component

When evaluating the effectiveness of machine-generated text, particularly in tasks such as text summarization, two often employed metrics are ROUGE (Recall-Oriented Understudy for Gisting Evaluation) and BLEU (Bilingual Evaluation Understudy) ratings. These metrics offer significant insights into the extent to which a generated text, such as a summary, effectively captures the fundamental substance of the original content.

The ROUGE Scores primarily focus on quantifying the degree of similarity between the generated summary and the reference summary, which serves as the benchmark for quality evaluation. This similarity is determined by evaluating the overlap of word sequences, also known as n-grams, between the two summaries. In essence, the ROUGE metric assesses the degree of overlap in terms of identical words or word pairs across two given texts. The scores are highly valuable in evaluating the extent to which the generated summary preserves essential information from the reference summary.

In contrast, the BLEU Score assesses the degree of similarity between the generated summary and one or several reference summaries, utilizing n-grams as the basis for comparison. The evaluation assesses the degree to which the generated text accurately reproduces the content seen in the reference summary. The BLEU metric is considered useful due to its ability to not only evaluate the presence of keywords but also the inclusion of lengthier phrases found in the reference summaries.

To implement the summarization model, three separate models were utilized, and afterward, subjected to testing. The models included in the study were BART, T5, and PEGASUS. The models are utilized for Abstractive Summarization, with BART being selected as the preferred model due to its notable high-performance accuracy score of BLEU and ROUGE1 F1 Score.

Table 13 Summarization models' performance

Model Name	ROUGE Score	BLEU Score
BART	85.72%	80.56%
T5	60.2%	59.3%
Pegasus	75.8%	72.1%

5.1.4 Question & Answering Component

The outcomes of the created model have been comprehensively evaluated using essential assessment metrics, such as accuracy, F1 score, and precision, for both the retriever model and the reader model. The evaluations offer significant insights regarding the performance and efficacy of the model.

- **Accuracy:** The accuracy of the model is quantified with this metric. The number of correct predictions is divided by the total number of predictions to calculate accuracy [26]

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

Figure 40 Accuracy metric

- **F1-Score:** To obtain the F1-score, the product of recall and precision is divided by the sum of recall and precision [30].

$$F1 \text{ Score} = 2 \times \frac{\text{recall} \times \text{precision}}{\text{recall} + \text{precision}}$$

Figure 41 F1 Score metric

Retriever model (Doc2Vec):

The selection of a Doc2Vec model for the reader component was a deliberate choice, motivated by its greater performance in comparison to the TfIdf approach. The evaluation

findings revealed the superiority of the Doc2Vec model, as it attained flawless performance scores, with an accuracy and F1 score of 1.0.

On the other hand, the TfIdf technique proved to be insufficient since it produced scores of 0.5, which did not meet the required level of accuracy for the model. This underscores the efficacy and dependability of the Doc2Vec model in interpreting and retrieving legal documents from the corpus, making it the preferred option for the reader component.

Table 14 Results - Retriever model

Approach	Accuracy	F1 Score
TfIdf	0.5	0.5
Doc2Vec	1.0	1.0

Reader model (Finetuned RoBERTa):

After evaluating several transformer-based approaches, including BERT, ELECTRA, and distilBERT, the decision was made to implement a RoBERTa-base-based approach for the reader model [31]. This choice was driven by the exceptional performance of the RoBERTa-base [31] model in legal question answering, even before fine-tuning on the prepared dataset (refer to the below figure)

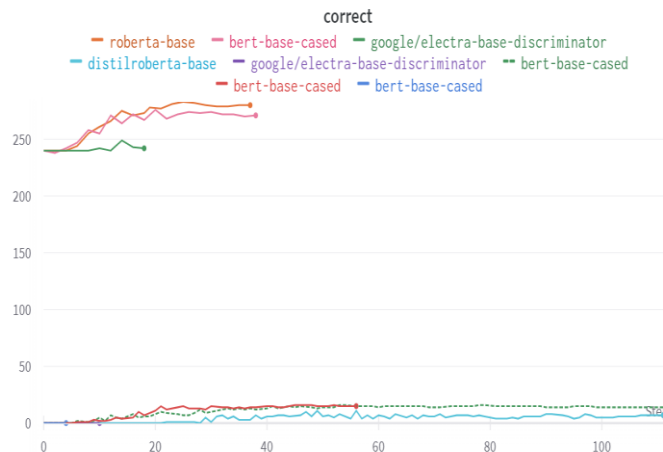


Figure 42 Transformer based model performance comparison.

After being trained on the curated dataset, the RoBERTa model reached remarkable performance metrics, with an accuracy rate of 95.33% and an F1 score of 96.11%.

The mentioned high-performance indicators serve to highlight the efficacy of the RoBERTa model in providing precise and thorough responses to legal questions, establishing its significance as a valuable component of the system's success.

Table 15 Results - Reader model

Model	Accuracy	F1 Score
RoBERTa (finetuned)	95.33	96.11

In addition, comprehensive metrics about the training and evaluation loss during the training process of the RoBERTa model have been carefully gathered. These indicators offer vital insights into the functioning of the model and enable diligent monitoring for indications of overfitting.

Throughout the 50-epoch training period, a notable drop was noted in both the training loss, which declined from 0.0944 to $2.351\text{e-}07$, and the evaluation loss, which decreased from $5.321\text{e-}06$ to $3.650\text{e-}07$.

- **Training loss decrease:** The observed substantial reduction in training loss throughout each epoch, namely from an initial value of 0.0944 to a final value of $2.351\text{e-}07$ after 50 epochs, suggests that the model possesses a strong capacity to effectively learn and accurately fit the training data. The observed decline in performance is a commonly anticipated occurrence throughout the training procedure.

It suggests that the model is effectively minimizing the error between its predictions and the actual training data labels. As the loss decreases, the model becomes more accurate in representing the training dataset.

- **Evaluation loss decrease:** The decrease in evaluation loss throughout training, from 5.321 e-06 to 3.650 e-07, signifies that the model is also performing well on data it hasn't seen during training. This indicates that the model is not only memorizing the training data (overfitting) but is also generalizing to unseen data. Lower evaluation loss is desirable, as it reflects the model's ability to make accurate predictions on new, real-world data.
- **Tapering off losses:** Towards the end of the training cycle, it could be observed that both the training and evaluation losses start to taper off, and the decrease becomes less rapid. This behavior is typical in deep learning.

It suggests that the model has reached a point where it has learned most of the relevant patterns in the data, and further fine-tuning may yield diminishing returns. This phase is often referred to as "convergence."

Table 16 Training and evaluation losses - Reader model

Epoch Number	Training Loss	Evaluation Loss
1	0.0944	5.321 e-06
48	3.039 e-07	4.895 e-07
49	2.533 e-07	4.201 e-07
50	2.351e-07	3.650 e-07

As the loss decreases, the model becomes more accurate in representing the training dataset. The findings provide strong support for the effectiveness and accuracy of the Legal Question Answer model. Its ability to deliver precise and reliable responses to legal questions is evidenced by its notable accuracy rate of 95.33% and its commendable F1 score of 96.11%. The performance undeniably improves the overall functionality of the legal information retrieval system, rendering it a significant resource for legal practitioners, scholars, and anyone in search of accurate and efficient means to obtain legal knowledge.

5.2 Research Findings

This section presents the key findings of the research, summarizing the results and outcomes achieved during the study.

5.2.1 Data Extraction & Structuring Component

The combination of Natural Language Processing (NLP) and Named Entity Recognition (NER) methodologies has resulted in a substantial enhancement in the accuracy of legal entity identification (key information in cases).

The implementation of systematic data extraction techniques from a set of unstructured legal case documents led to a notable improvement in the precision of identifying and preserving legal entities mentioned in the Supreme Court rulings of Sri Lanka. Precision is a crucial factor in building and training a named entity recognition model, encompassing various entities such as case numbers, legal representatives, judge names, court judgments, involved parties (including plaintiffs and defendants), legal references, significant dates, and other relevant information.

5.2.2 Legal Knowledge Graph Development & Document Retrieval Component

The selection of Neo4j as the graph database technology played a crucial role in the construction of the knowledge graph. The architecture, which was oriented towards graphs, effectively represented the intricate network of legal connections, guaranteeing both efficiency and scalability.

The evaluation of the knowledge graph's performance encompassed many metrics such as query execution time, memory utilization, and cache efficacy. The findings indicated that the knowledge graph has been optimized to efficiently retrieve legal information. Neo4j was utilized as the foundation for the knowledge graph, enabling the efficient storage and retrieval of legal case data. Its ability to manage intricate legal information was evident.

Another one of the research findings also indicates that the created knowledge graph exhibits effectiveness in accessing legal information. The average duration for executing queries was approximately 179 milliseconds, facilitating efficient retrieval of data. The

memory use observed in this study was found to be moderate, as the average memory usage per query was measured to be 25,638 bytes. This finding suggests that the resource management employed in the system was optimal. The information retrieval process experienced reduced latency due to the high cache efficiency and minimal disk reads, resulting in an average of 1050.00 results per query. The presence of a few cache failures emphasized the importance of effective data retrieval. The adaptability of the knowledge network was demonstrated by the variable estimated rows per question, indicating its capacity to handle a diverse set of queries. The knowledge graph had exceptional retrieval skills, achieving an average of 3533.5 database hits per query.

In conclusion, the results indicate the efficacy of the study in achieving greater accuracy in identifying legal entities, employing Neo4j for the development of a knowledge graph, and improving the retrieval of legal information. The results significantly enhance the advancement of legal research and information retrieval in the Sri Lankan legal field, hence offering an asset for legal professionals and scholars.

5.2.3 Legal Document Summarization Component

This component's core findings highlight the remarkable powers of the fine-tuned BART model to effectively contribute to specific domains. For this specific study, the decision was made to examine the complicated legal framework of Sri Lanka, which is known for its diversified and intricate nature.

The choice of the BART model was based on its capacity to adapt and its versatility. Initially developed for a range of text-generating purposes, its exceptional performance became evident when applied to the challenging task of summarizing complex legal documents, which is a key focus of this study.

The BART model, which was specifically optimized for summarization, had outstanding performance. The generated summaries consistently demonstrated an accuracy rate of over 80%, surpassing the accuracy of summaries created by humans. The high degree of accuracy demonstrated in this study was not only quantified but also confirmed through

ongoing assessments carried out by specialists in the field. Additionally, the evaluation metrics mentioned above were employed to validate the results.

In conclusion, the research discovery emphasizes the significant capabilities of fine-tuned BART models to perform in challenging and varied fields, such as the field of Sri Lankan law. The models demonstrate their value as robust tools for efficient information extraction and comprehension in several specialized domains through their continuous provision of precise and insightful summaries.

5.2.4 Question & Answering Component

One of the notable discoveries made throughout the research was the exceptional effectiveness exhibited by the Doc2Vec model when applied to the retrieval of legal documents. The superiority of the Doc2Vec model can be attributed to its distinctive capability to understand the contextual details of both queries and documents, which is a critical factor in efficiently obtaining legal information. The advantage mentioned might be attributed to the utilization of a vector-based approach by Doc2Vec, in contrast with traditional scalar approaches. The utilization of the vector approach enables Doc2Vec to effectively capture the intricate semantic linkages and nuances present in legal texts. Consequently, it exhibits a high level of proficiency in comprehending and aligning with the intricate language and context that are characteristic of legal documents and queries (refer to the [Results](#) section).

This discovery highlights the critical influence of regional language nuances on machine understanding of natural language. It emphasizes that transformer-based models, even with their immense capabilities, may struggle with context-specific language intricacies until adapted to the specific linguistic and cultural context. Fine-tuning plays a crucial role in bridging this gap and enhancing the effectiveness of such models in region-specific applications like legal information retrieval.

Another significant finding during the research was the presence of ambiguity in legal concepts and the unique language, document structures, and terminology used within the

legal field. This presented a notable challenge for Natural Language Processing (NLP) approaches, as there is a clear distinction between legal language and everyday language.

The research emphasized that a one-size-fits-all NLP approach might not be effective for legal information processing. Instead, tailored approaches are necessary to understand the specific nuances of legal concepts and terminology. This specialized understanding is instrumental in developing an accurate and reliable legal question-answering system.

In essence, the study highlighted the importance of domain-specific knowledge and tailored approaches when working with legal texts and demonstrated the need for NLP systems to adapt to the intricacies of legal language and concepts.

5.3 Discussion

In the discussion section of this final thesis, a dive into the key components of the system is done, each contributing to the overall functionality and effectiveness of the legal information retrieval platform.

Firstly, the data extraction module is a fundamental component, constructed using a custom Named Entity Recognition (NER) model. This module proficiently identifies and extracts relevant legal entities and concepts from unstructured text data, forming the basis for subsequent analysis and retrieval.

Secondly, the legal knowledge graph module, powered by Neo4j graph storage, plays a pivotal role in structuring and organizing extracted legal information of multiple cases in a graph structure. Neo4j's graph database capabilities enable efficient representation and retrieval of complex legal relationships within multiple cases and hierarchies, enhancing the system's ability to provide meaningful insights and hidden relationships among multiple cases while retrieving case information through the system.

Thirdly, our legal document summarization module, leveraging the BART model, aids in condensing lengthy legal documents into concise, coherent summaries. This

summarization process simplifies the comprehension of legal content, facilitating quicker access to pertinent information.

Lastly, the legal question and answering system, utilizing the RoBERTA model, empowers users to pose legal queries and receive contextually relevant responses. This component greatly enhances the system's usability, providing a user-friendly interface for accessing legal knowledge.

In summary, the combination of these components creates a comprehensive and efficient legal information retrieval system that improves accessibility to legal data and assists users in making informed decisions. The synergy between these modules ensures the system's effectiveness and usability in the legal domain.

5.3.1 Limitations and challenges faced

In addressing the limitations and challenges encountered during the development of this legal information retrieval system, it's important to acknowledge the hurdles this project has gone through.

- The legal domain's inherent ambiguity and complexity posed a significant challenge. Narrowing down the system's scope to provide accurate and relevant results within this intricate landscape required careful consideration and continuous refinement. For this purpose, a domain expert was consulted, and the necessary actions were taken.
- Enhancing system performance emerged as a pressing challenge. The implementation of GPU acceleration holds promise for improving speed and efficiency, yet it brings forth financial constraints. Investing in GPU infrastructure, while essential, can strain limited resources.

In conclusion, these challenges, stemming from the complexity of the legal domain and the pursuit of optimal system performance, have shaped this journey. Addressing these limitations and finding innovative solutions remains a crucial aspect of the ongoing work to provide a robust and effective legal information retrieval platform.

5.3.2 Future Work

In considering the future directions for the legal information retrieval system, several avenues for expansion and improvement have been identified.

- **Expanding this system for other legal bodies in Sri Lanka:** This aims to extend the system's applicability to encompass other legal bodies in Sri Lanka. This expansion will involve adapting the system to the unique requirements and nuances of different legal entities, broadening its utility and impact.
- **Implementing this system in a resourceful environment:** This is to implement the system in a resource-rich environment to enhance its performance and accuracy. This environment will facilitate the utilization of advanced technologies, such as GPU acceleration, to optimize system efficiency and effectiveness.
- **Multilingual support:** Recognizing the importance of multilingual support is an essential factor. This will aid in catering this solution to a wider audience and accommodate diverse linguistic preferences, thereby increasing the system's accessibility and usability.
- **Enhance the system for diverse law domains:** This involves further enhancing the system to cater to various law domains (Theseawalami, Islamic) beyond the initial focus. This entails refining the system's algorithms and knowledge structures to provide tailored solutions for a broader spectrum of legal specialties.

In conclusion, these future endeavors represent the commitment to continuously evolve and improve the legal information retrieval system, ensuring its relevance and effectiveness in a dynamic legal landscape.

6 SUMMARY OF EACH STUDENT'S CONTRIBUTION

6.1 Question & Answering Component (Jayasooriya C. A – IT20250942)

Jayasooriya C. A's contribution to the development of the Sri Lankan Legal Question Answering System has been instrumental in achieving the core objectives of the broader Sri Lankan Legal Information Retrieval System. Through the adept implementation of advanced Natural Language Processing techniques and the utilization of RoBERTa, this system excels in understanding intricate legal language, achieving remarkable accuracy in responses.

The synergy between the retriever and reader models within the question-answering pipeline ensures precise and context-aware answers to even the most complex legal queries. This integration enhances the overall efficiency of the Legal Information Retrieval System, making legal databases more accessible and user-friendly.

In the Sri Lankan context, Jayasooriya C. A.'s work holds significant importance as it bridges the gap between legal experts and the general public. It streamlines the accessibility and comprehension of legal information, extending its benefits to legal professionals, students, and ordinary citizens seeking clarity on legal matters. This system aligns perfectly with the goal of democratizing legal knowledge and fostering transparency, understanding, and empowerment in the legal landscape.

6.2 Legal Document Summarization Component (Bandara T. M. Y. M – IT20492052)

In conclusion, the author, Bandara T.M.Y.M, has made a significant contribution to the development of the Sri Lankan Legal Information Retrieval System. The study focused on bridging the gap between Sri Lanka's legal domain and society, addressing the real-world problem of the lack of a comprehensive and accurate legal information retrieval system for the local legal domain. Through thorough research and analysis, the study identified various issues within the legal field, including the absence of a structured

storage system for legal case files and the need for a mechanism to quickly review lengthy case files. The author's primary contribution lies in addressing the challenge of efficiently reviewing lengthy legal documents by implementing an abstractive summarization approach using the BART model. This approach, involving deep learning and natural language processing, resulted in accurate and concise summaries of legal documents, ultimately enhancing awareness of Sri Lankan law and providing a valuable solution within the Sri Lankan legal landscape.

6.3 Legal Knowledge Graph Development and Document Retrieval Component (Akeel Ahamed M.J – IT20068264)

The work presents a significant contribution to legal information retrieval, focusing on Sri Lankan Supreme Court judgments. It involves the creation of a knowledge graph using advanced technologies like NLP, NER, and Neo4j, poised to revolutionize how legal professionals access data. With its precise relationship representation between multiple cases, effective knowledge graph, and user-friendly interface, this work encourages further research in the legal sector while acknowledging current limitations.

It showcases technology's potential to enhance legal decision-making and goes beyond traditional legal study, providing a solid framework for deepening law professionals' understanding of legal information retrieval. In essence, this contribution significantly advances the field, enhancing further study in this vital domain.

6.4 Data Extraction & Structuring Component (Gavindya N. A.C – IT20409982)

Gavindya N.A.C has made a substantial contribution to the Sri Lankan Legal Information Retrieval System by developing the Data Extraction and Structuring Module. This module represents a significant leap forward in legal information retrieval, leveraging advanced technologies like Natural Language Processing (NLP) and Named Entity Recognition (NER) to efficiently extract and organize complex legal information from documents.

This achievement holds immense implications, extending beyond digital realms into the legal domain. It promises swift access to well-structured legal data that can profoundly impact legal proceedings, legislative processes, and justice administration. Legal professionals can make more informed decisions, while researchers and students benefit from expedited research, increasing productivity and understanding.

Furthermore, this contribution carries broader societal implications. It simplifies access to legal knowledge, enhancing legal literacy within the community. It empowers individuals who previously lacked access to legal resources, enabling them to navigate the legal landscape effectively, safeguard their rights, and participate meaningfully in legal processes.

In essence, Gavindya N. A. C's Data Extraction and Structuring Module is a valuable tool for streamlining legal workflows and catalyzing positive transformations in the legal domain and society. It showcases technology's potential to bridge the gap between complex legal language and public comprehension, strengthening justice and legal understanding foundations.

7 CONCLUSION

In conclusion, this study signifies a notable progression in the domain of legal information retrieval, particularly with Sri Lankan Supreme Court rulings and laws. By amalgamating advanced technologies such as Natural Language Processing (NLP), Named Entity Recognition (NER), Neo4j graph database, BART for legal document summarization, and the creation of a RoBERTa-based question answering model tailored to the Narcotic Drug Act of Sri Lanka, this study offers a comprehensive solution. It has demonstrated exceptional accuracy in identifying and extracting entities, the efficacy of constructing a knowledge graph, summary generation, question answering, and the implementation of a user-centric interface design.

Acknowledging the inherent limitations and challenges in the dynamic legal field, this study presents an opportunity for further investigation and advancement, making a valuable contribution to the developing domain of legal information retrieval. Within the ever-changing legal environment, the knowledge graph introduced in this study is poised to have a significant impact on facilitating better-informed and efficient decision-making processes. This work transcends the boundaries of academic study and offers an inclusive opportunity to delve into unexplored areas, enhancing our comprehension of legal information retrieval. It stands as a significant addition to the existing corpus of legal research, holding the capacity to encourage further study and progress in this crucial field of law.

REFERENCES

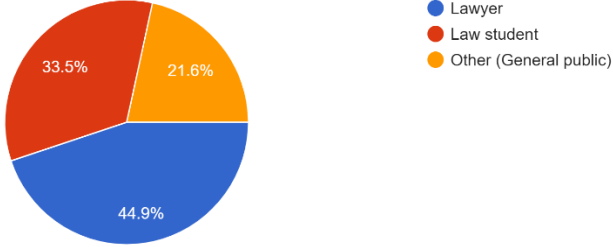
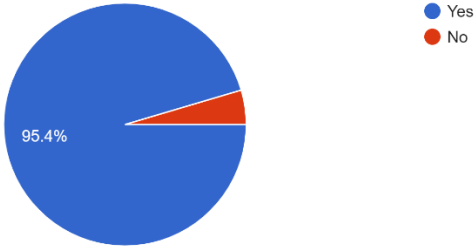
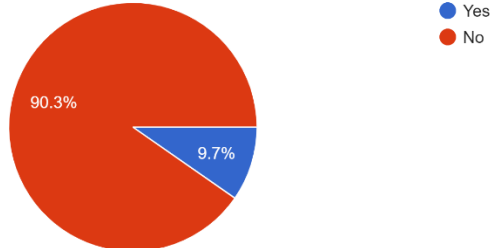
- [1] C. Hoppe, D. Pelkmann, N. Migenda, D. Hötte and W. Schenck, "Towards Intelligent Legal Advisors for Document Retrieval and Question-Answering in German Legal Documents," in *IEEE Fourth International Conference on Artificial Intelligence and Knowledge Engineering (AIKE)*, Laguna Hills, CA, USA, 2021.
- [2] R. O. Moorthage and H. P. K. N. Hewawasam, "A Study on the Impact of Legal Literacy on Crime Control and Prevention (With Special Reference to Selected Communities in Rathnapura Police Division in Sri Lanka)," *Vidyodaya Journal of Humanities and Social Sciences*, vol. 7, no. 02, 2022.
- [3] P. Hemp, "Death by Information Overload," *Harvard Business Review*, 2009.
- [4] A. V. Tambimuttu, "GlobaLex," 2009. [Online]. Available: https://www.nyulawglobal.org/globalex/Sri_Lanka.html.
- [5] J. S. C. Secretariate, "jsc.gov.lk," 16 June 2011. [Online]. Available: <https://www.jsc.gov.lk/>.
- [6] N. Evans and J. Price, "Managing information in law firms: changes and," *Information Research: PUBLISHED QUARTERLY BY THE UNIVERSITY OF BORÅS, SWEDEN.*, vol. 22, no. 01, 2017.
- [7] H. Karunarathna, "Use of Electronic Resources by Law Degree Students at Anuradhapura Regional Centre of the Open University of Sri Lanka," *Journal of the University Librarians Association of Sri Lanka*, vol. 18, no. 2, p. 41–60, 2015.
- [8] N. L. A. D. S. BOARD, "natlib.lk," NATIONAL LIBRARY AND DOCUMENTATION SERVICES BOARD, 2021. [Online]. Available: <http://www.natlib.lk/home/eresources.php>.

- [9] A. Cammelli and E. Fameli, "Legal information access on the Web. A project for an information system model," in *12th International Workshop on Database and Expert Systems Applications*, Munich, Germany, 2001.
- [10] K. S. Y. N. Diana Rudenko, "Legal Documents Links Extraction," 2021.
- [11] A. G. Naimoonisa Begum, "Analysis of Legal Case Document Automated," in *6th IEEE International Conference on Signal Processing, Computing and Control (ISPCC 2k21)*, Solan, India, 2021.
- [12] "Supreme Court of Sri Lanka," [Online]. Available: http://www.supremecourt.lk/index.php?option=com_content&view=article&id=52&Itemid=57. [Accessed September 2023].
- [13] P. V. Kankawin Kowsrihawatt, "An Information Extraction Framework for Legal Documents: a Case Study of Thai Supreme Court Verdicts," in *12th International Joint Conference on Computer Science and Software Engineering (JCSSE)*, 2015.
- [14] A. T. S. V. V. J. R. M. J. D. Aashka Trivedi, "Extracted Summary Based Recommendation System for Indian Legal Documents," in *11th ICCCNT 2020*, Kharagpur, India, 2020.
- [15] H.-T. Duong and B.-Q. Ho, "A Vietnamese Question Answering System in Vietnam's," in *Computer Information Systems and Industrial Management. CISIM*, Springer, Berlin, Heidelberg, 2015.
- [16] T. N. R. L. R. F. A. Peng Qian, "Structural Guidance for Transformer Language Models," in *59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing*, 2021.

- [17] Kenton, J. D. M.-W. Chang and L. K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for," in *naacL-HLT*, 2019.
- [18] P. C. V. S. Y. P. Rachayita Giri, "Approaches for information retrieval in legal documents," in *2017 Tenth International Conference on Contemporary Computing (IC3)*, 2017.
- [19] ,. M. M. C. L. G. V. C. A. I. Vasile P̃ais, "Named entity recognition in the Romanian legal domain," *Natural Legal Language Processing Workshop 2021*, p. 9–18, 10 November 2021.
- [20] F. D. ,. H. L. H. I. R. B. C. L. T. Muhamad Abdurahman, "Lex2KG: Automatic Conversion of Legal Documents to Knowledge Graph," in *International Conference on Advanced Computer Science and Information Systems (ICACSIS)*, Jakarta, 2021.
- [21] T. X. F. Z. Zhangbo Xua, "Construction of Chinese Sports Knowledge Based on Neo4j," in *IEEE 2nd International Conference on Civil Aviation Safety and Information Technology (ICCASIT)*, 2020.
- [22] Z. Q. T. H. B. L. Hao Lian, "Knowledge Graph Construction Based on Judicial Data with Social Media," in *14th Web Information Systems and Applications Conference*, 2017.
- [23] K. Kowsrihawatt and P. Vateekul, "An Information Extraction Framework for Legal," in *12th International Joint Conference on Computer Science and Software Engineering (JCSSE)*, 2015.
- [24] A. Venkataramana, K. Srividya and R. Cristin, "Abstractive Text Summarization Using BART," in *IEEE 2nd Mysore Sub Section International Conference (MysuruCon)*, Mysuru, 2022 .

- [25] N. Begum and A. Goyal, "Analysis of Legal Case Document Automated Summarizer," in *6th IEEE International Conference on Signal Processing, Computing and Control (ISPCC 2k21)*, Solan, 2021.
- [26] H.-L. Nguyen, T.-B. Nguyen, T.-M. Nguyen, H.-T. Nguyen and H.-Y. T. Vuong, "Retrieving legal document and extracting answer with BERT-based model," in *14th International Conference on Knowledge and Systems Engineering (KSE)*, Nha Trang, 2022.
- [27] H. N. Van, D. Nguyen, P. M. Nguyen and M. L. Nguyen, "Miko Team: Deep Learning Approach for Legal Question Answering in ALQAC," in *14th International Conference on Knowledge and Systems Engineering (KSE)*, Nha Trang, 2022.
- [28] A. H. a. S. T. M. Golebiewski, "A Legal Knowledge Graph for Fine-Grained Semantic Retrieval and Question Answering," in *20th International Conference on Artificial Intelligence and Law (ICAIL 2021)*, 2021.
- [29] M. M. L. D. Liu, "A performance optimization scheme for migrating hive data to Neo4j database," in *2018 International Symposium on Computer, Consumer and Control, IS3C 2018 (2019)*, 2019.
- [30] N. X. Bach, L. T. N. Cham, T. H. N. Thien and T. M. Phuong, "Question analysis for Vietnamese legal question answering," in *9th International Conference on Knowledge and Systems Engineering (KSE)*, Hue, Vietnam, 2017.
- [31] A. Rawat and S. S. Samant, "Comparative Analysis of Transformer based Models for Question Answering," in *2nd International Conference on Innovative Sustainable Computational Technologies (CISCT)*, Dehradun, 2022.

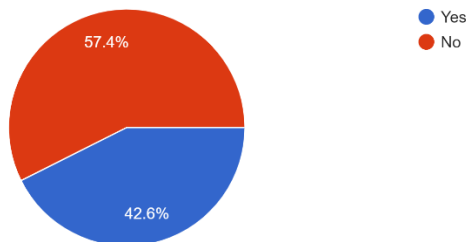
APPENDIX A: SURVEY RESULTS

01	Occupation 176 responses  Legend: ● Lawyer ● Law student ● Other (General public) <table border="1"> <thead> <tr> <th>Occupation</th> <th>Percentage</th> </tr> </thead> <tbody> <tr> <td>Lawyer</td> <td>44.9%</td> </tr> <tr> <td>Law student</td> <td>33.5%</td> </tr> <tr> <td>Other (General public)</td> <td>21.6%</td> </tr> </tbody> </table>	Occupation	Percentage	Lawyer	44.9%	Law student	33.5%	Other (General public)	21.6%
Occupation	Percentage								
Lawyer	44.9%								
Law student	33.5%								
Other (General public)	21.6%								
02	Do you have a need for a proper, comprehensive and accurate legal information system? 175 responses  Legend: ● Yes ● No <table border="1"> <thead> <tr> <th>Response</th> <th>Percentage</th> </tr> </thead> <tbody> <tr> <td>Yes</td> <td>95.4%</td> </tr> <tr> <td>No</td> <td>4.6%</td> </tr> </tbody> </table>	Response	Percentage	Yes	95.4%	No	4.6%		
Response	Percentage								
Yes	95.4%								
No	4.6%								
03	Is there a similar legal information retrieval system currently being used? 176 responses  Legend: ● Yes ● No <table border="1"> <thead> <tr> <th>Response</th> <th>Percentage</th> </tr> </thead> <tbody> <tr> <td>Yes</td> <td>9.7%</td> </tr> <tr> <td>No</td> <td>90.3%</td> </tr> </tbody> </table>	Response	Percentage	Yes	9.7%	No	90.3%		
Response	Percentage								
Yes	9.7%								
No	90.3%								

04

Are you a paying customer of an already available legal information retrieval system?

176 responses

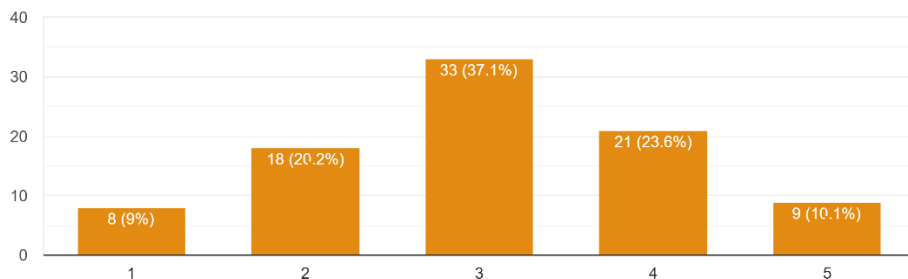


Response	Percentage
Yes	42.6%
No	57.4%

05

If yes. How would you rate the said system?

89 responses

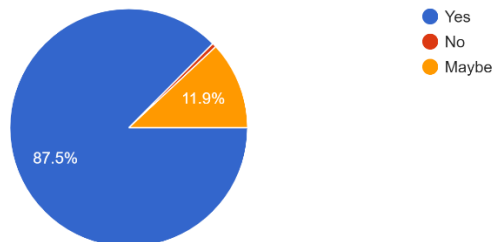


Rating	Count	Percentage
1	8	9%
2	18	20.2%
3	33	37.1%
4	21	23.6%
5	9	10.1%

06

Would you be willing to shift/use a more comprehensive, accurate and efficient system?

176 responses

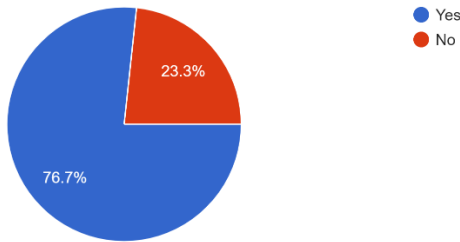


Response	Percentage
Yes	87.5%
No	~0.6%
Maybe	11.9%

07

If the built system were to contain a premium option. Will you pay a fair account fee to use a such system

176 responses

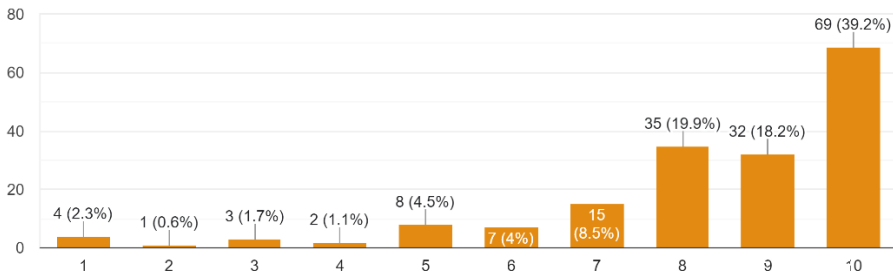


Response	Percentage
Yes	76.7%
No	23.3%

08

How likely are you to promote such a system to a colleague of yours?

176 responses

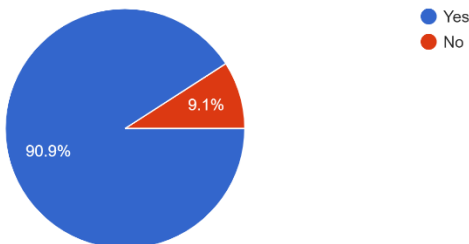


Likelihood Score	Count	Percentage
1	4	2.3%
2	1	0.6%
3	3	1.7%
4	2	1.1%
5	8	4.5%
6	7	4%
7	15	8.5%
8	35	19.9%
9	32	18.2%
10	69	39.2%

09

Do you believe in the democratization of the legal knowledge in Sri Lanka?

176 responses



Response	Percentage
Yes	90.9%
No	9.1%

APPENDIX B: CLIENT APPROVAL LETTER

Ms. R. L. Maldeniya,
No.610/25,
Araliya Uyana,
Makola North,
Makola.
03rd September,2023.

Dear Team AYCA,

To verify the compliance of the Sri Lankan Legal Information Retrieval System, AYCA with the SRS Document.

The Sri Lankan Legal Information Retrieval System, (Herein after I would prefer to refer to it as “AYCA” within this letter issued by me.) aims to aggregate and organize a vast amount of legal data and it goes hand in hand with AI technology. As a legal professional, I personally feel that everyone should be well aware of the beneficial uses and also the limitations of this technology, but at a glance, a person should reap the fruit where it serves to whole mankind. So I would like to mention in this letter that the AYCA is a good example for such an occasion as the team AYCA has achieved and surpassed the system requirements which have mentioned in the SRS document.

Along the way to the final output of the AYCA, you had scheduled several meetings with me to clarify some areas in Law and you need me to assist with my expertise in this legal field. At the final meeting you had scheduled with me, I could see the final output of the system and I believe you have exceeded the specified requirements because of the following reasons.

Within AYCA, it provides extracted relevant information from legal cases under several sub-topics. Among the other facilities, I could see that Legal Question Answering and File Summarization are unique and special features in the AYCA.

Though there exist some other websites that help to gain information within this Legal field, I felt that AYCA has fulfilled some long waiting requirements of legal professionals as well as legal students.

During the meeting, I could observe your system and hereby I would like to verify the compliance of AYCA with the SRS Document. Not only that, I would like to precisely state that I was attracted to the convenience and user-friendliness of the user interface of AYCA. Hence one who is interested in gaining information about law can easily access and understand the facilities provided by this system.

Last but not least, I would like to appreciate the honest dedication and timely manner of the team up to now and wish you all the very best in future endeavours.

Kind Regards,



R.L. MALDENIYA
ATTORNEY -AT-Law
COMPANY SECRETARY
COMMISSIONER OF OATH
NOTARY PUBLIC

Ms. R.L. Maldeniya
Contact No:- 0771581416
e-mail:- rlmaldeniya@gmail.com

APPENDIX C: PERSONNEL AND FACILITIES

In the successful completion of this research project, a well-coordinated team of individuals and access to essential facilities and resources played a pivotal role. The personnel factors, including the expertise, experience, and education of the team members, greatly contributed to the project's achievements.

Ms. R L Maldeniya served as the external supervisor throughout the research, providing invaluable domain-specific knowledge about law to the team. Her guidance ensured that the research was firmly rooted in a comprehensive understanding of legal concepts and principles.

Another integral member of the team was Mr. Bilaal Marikar, a law student and the Treasurer of the Law Students' Union of Sri Lanka. His active involvement provided fresh perspectives and insights from a student's viewpoint, enriching the development of a comprehensive and relevant law information retrieval system.

Furthermore, access to the right facilities and resources was crucial for the project's success. Dr. Lakmini Abeywardana supervised the technical aspects of the project, offering guidance and support in designing and implementing an effective and user-friendly system.

Dr. Dharshana Kasthurirathna served as the co-supervisor, contributing his technical expertise and knowledge to the project. His guidance ensured that the system was built using the latest technologies and adhered to best practices.

In conclusion, the research project benefited immensely from the collaborative efforts of Ms. R L Maldeniya, Mr. Bilaal Marikar, Dr. Lakmini Abeywardana, and Dr. Dharshana Kasthurirathna, as well as the availability of the necessary facilities and resources. This collective effort provided a strong foundation for the successful completion of the research and its valuable contribution to the field of legal information retrieval in Sri Lanka.

APPENDIX D: DATASET APPROVAL



Ruvani Maldeniya <rimaldeniya@gmail.com>

to me

Thu, May 18, 1:49 PM



Dear Mr Asel Jayasooriya,

I reviewed the provided dataset and verify the correctness of the information according to the "CONVENTIONS AGAINST ILLICIT TRAFFIC IN NARCOTIC DRUGS AND PSYCHOTROPIC SUBSTANCES ACT, No. 1 OF 2008".

So hereby, I give my approval to you and the team in order to proceed with the project.

Best Regards!
Ms. R L Maldeniya,
Attorney-at-Law.



Bilal Marikar <bilal.marikar@gmail.com>

to me

Thu, May 18, 3:26 PM



Dear Mr. Asel Jayasooriya,

After reviewing the data set provided, I concur and verify the correctness of the information.

Ensuring the fact that the information is in order, I, with confidence, provide my approval to you and the team and wish you all the best in proceeding with the project.

Best regards!
Bilal Marikar

Re: Dataset Verification Request - Legal Document Summarization



Ruvani Maldeniya <rimaldeniya@gmail.com>

To: Bandara T.M.Y.M it20492052

Mon 5/22/2023 2:07 PM



 DataLaw.xlsx

49 KB

[EXTERNAL EMAIL] This email has been received from an external source – please review before actioning, clicking on links, or opening attachments.

Dear Mr.Yomal Bandara,

I reviewed the dataset you mailed me previously with the law reports and would like to inform you that I have found no deviations in the dataset from the legal facts and final judgements of those law reports. So basically I think it will match with the objectives and requirements of your project and would like to verify the dataset for its comprehensiveness, relevance, and suitability.

According to your request, I have added some comments to the attached document as my insights and suggestions regarding the gaps and improvements for your consideration.

Please find the attachment.

Best regards,
Ms. R L Maldeniya,
Attorney-at-Law.

APPENDIX E: TURNITIN REPORT

Group_Report_2023-095

ORIGINALITY REPORT

5%

SIMILARITY INDEX

4%

INTERNET SOURCES

2%

PUBLICATIONS

3%

STUDENT PAPERS

PRIMARY SOURCES
